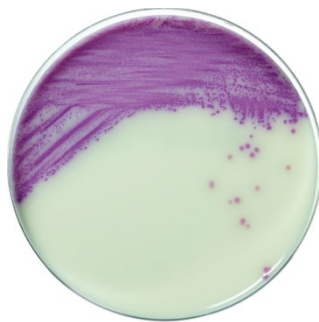


RAPID'*Salmonella* Agar

User Guide

Chromogenic media for the detection of *Salmonella* spp. in the analysis of food products for human consumption, animal feed, and in environmental samples

Catalog #3563961, Prepared plates, 90 mm x 20 dishes
Catalog #3563963, Prepared plates, 90 mm x 120 dishes
Catalog #3564705, Dehydrated, 500 g



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Table of Contents

Section 1	Introduction	1
Section 2	RAPID' <i>Salmonella</i> Principle	1
Section 3	Theoretical Formula	1
Section 4	Shelf Life and Storage	1
Section 5	Materials Required but Not Supplied	1
	Equipment.....	2
	Supplies	2
Section 6	Precautions, Limitations of Use, and Quality Control	2
Section 7	Protocol.....	3
	Preparation of Dehydrated Medium.....	3
	Preparation of Selective Supplement	4
	Sample Enrichment	4
	Inoculation and Plate Reading.....	5
Section 8	Confirmation of Positive Results.....	5
Section 9	Confirmation of Other Methods.....	6
Section 10	Test Performance and Validations	6
Section 11	References.....	7
Section 12	Revision History.....	7

Section 1

Introduction

Salmonella is a gram-negative, non-spore-forming, rod-shaped bacterium that is predominantly motile, with only a few exceptions. *Salmonella* has been one of the two most frequently reported causes of bacterial foodborne illness. Common symptoms of salmonellosis include diarrhea, fever, and abdominal cramps occurring 12–72 hr after infection. Studies have shown that for each laboratory-confirmed case of *Salmonella*, 38 cases occur in the community that remain unreported. Testing for the presence of *Salmonella* in food and in the processing environment is an important practice. The use of chromogenic substrates in culture media increases the ease of use of the media with results based on color changing enzymatic reactions.

Section 2

RAPID'*Salmonella* Principle

RAPID'*Salmonella* Agar allows the presumptive identification of motile and non-motile *Salmonella* spp. by detecting C8-esterase activity. Simultaneous screening of β -glucosidase activity permits the differentiation of *Salmonella* colonies from those of other *Enterobacteriaceae*. After incubation, *Salmonella* appear as readily identifiable, typically magenta, colonies, whereas non-*Salmonella* bacteria grow as blue or colorless colonies. RAPID'*Salmonella* Agar permits detection of motile and non-motile *Salmonella*, as well as lactose-positive *Salmonella* Typhi and *Salmonella* Paratyphi.

Section 3

Theoretical Formula

Nutritive mix	14.5 g
Selective agents	14 g
Chromogenic mix	2.3 g
Agar	12.7 g
Distilled water	qsp 1,000 ml

Final pH at 25°C = 7.2 ± 0.2

Section 4

Shelf Life and Storage

- Dehydrated agar: 2–8°C in carefully sealed package and stored in a dry and dark place
- Prepoured agar: 2–8°C in a dark place
- Plate prepared from the dehydrated agar: 15 days at 2–8°C in carefully sealed package and stored in a dry and dark place

Section 5

Materials Required but Not Supplied

Equipment

- All usual laboratory equipment
- Hot plate
- Scale, sensitivity of 0.1 g
- Stirrer/homogenizer
- Thermostatically controlled incubator or incubation room, precise to $\pm 1^{\circ}\text{C}$
- Water bath

Supplies

- Enrichment medium: Buffered Peptone Water (BPW) (catalog #3554179, 225 ml x 6 bottles; 3564684, dehydrated 500 g; 3555790, 5 L x 2 bags; 3555795, 3 L x 4 bags)
- Inoculating loops
- PIF Supplement (catalog #12013322, 2 g bottle)
- RAPID'*Salmonella* selective supplement for short protocol (catalog #3564709, 10x concentrated x 100 capsules; 3564710, 1x concentrated x 100 capsules; 3564712, 1 box x 100 tests)
- Rappaport Vassiliadis Soy (RVS) for double enrichment protocol (catalog #3555773, 10 ml x 25 tubes x 25; 3564324, dehydrated 500 g)
- Sterile petri dishes
- Sterile pipets
- Sterile weigh bags

Section 6

Precautions, Limitations of Use, and Quality Control

Precautions

- Respect Good Laboratory Practice (EN ISO 7218). Appropriate protection, such as gloves and lab coats, should be worn when working with potentially infectious live bacteria such as *Salmonella*
- Media that have come in contact with food samples should be considered contaminated and should be disposed of in accordance with local rules and regulations

- If the whole selective supplement capsule is added to the BPW, sterile tweezers must be used to add the capsule to the bag. (We recommend that you check that the capsule actually opened during the stomaching stage.) If the capsule contents are handled with the fingers, it cannot be added to the enrichment broth due to the risk of contamination
- The RAPID'*Salmonella* Capsules and RAPID'*Salmonella* Supplement Tablets contain selective agents and an excipient. Selective agents dissolve very well. The excipient, however, remains in suspension and may create a deposit when the contents of the capsule are diluted in a small quantity of BPW or sterile distilled water. Always shake well, therefore, before using the concentrated solution

Limitations of Use

- ONPG confirmation test excludes confirmation of lactose-positive *Salmonella*
- Although the most prevalent *Salmonella* strains can be detected by *Salmonella* Confirm Latex Kit, the kit did not detect 41 of the 150 strains tested during the NF VALIDATION extension of RAPID'*Salmonella*
- Some strains of *Salmonella* (a few are part of *Salmonella* serovar Dublin and *S. bongori* species) can show a weak magenta color due to low esterase activity
- In the context of NORDVAL validation, the double enrichment protocol was validated with an RVS enrichment time of 24 ± 2 hr only
- After the incubation step, the plate can be stored at 2–8°C for 72 hr. Some non-*Salmonella* strains (for example, *E. coli*) can develop pink-colored colonies after this storage. These doubtful colonies give a negative confirmation test

Quality Control

- Every product manufactured and marketed by Bio-Rad is subject to a quality assurance procedure at all stages, from reception of raw materials through to marketing of the finished products. Each batch of finished product undergoes quality control according to EN ISO 11133 and is marketed only if it satisfies the acceptability criteria. Documentation relative to the production and quality control of each batch is kept on file
- For SDS product safety information and certificate of analysis, visit www.bio-rad.com

Section 7 Protocol

Preparation of Dehydrated Medium

1. Always shake bottle before use.
2. Dissolve 43.5 g of powder in 1 L of distilled water and mix until a homogenous suspension is obtained.
3. Heat gently, agitating frequently, then bring to a boil for less than 1 min.

4. Do not prolong heating. **Do not autoclave.**
5. Cool the medium to 50°C. Dispense in petri dishes and leave on the bench to dry.
6. One 500 g bottle of powder makes 11.5 L of medium.

Preparation of Selective Supplement

1. One capsule (3564710) is ready to use for 25 g or 25 ml of sample diluted in 225 ml of BPW. Either the whole capsule or its contents can be added to the sample bag before the mixing step. In order to make handling easier, we recommend opening the capsule and pouring out the contents.
2. Concentrated (10x) capsules (3564709) must be diluted before use. Pour the contents of one capsule into 10 ml of BPW or sterile distilled water. Homogenize to obtain a red concentrated solution. Use 1 ml of concentrated solution for 25 g or 25 ml of sample diluted in 225 ml BPW. For large sample size, adjust the selective supplement volume to the volume of enrichment mix to obtain concentration of 4:1000 g/ml. Diluted solution can be stored for 1 week at ambient temperature or at 2–8°C.
3. Boxed supplement (3564712) is concentrated and must be diluted before use. Open the box and fill with 100 ml of BPW or sterile distilled water. Homogenize by agitating vigorously in order to obtain a red concentrated solution. Use 1 ml of concentrated solution for 25 g or 25 ml of sample diluted in 225 ml BPW. For large sample size, adjust the selective supplement volume to the volume of enrichment mix to obtain concentration of 4:1000 g/ml. Diluted solution can be stored for 1 week at ambient temperature or at 2–8°C.
4. Open the bottle of PIF Supplement (12013322) and fill with 40 ml of a 1:1 ethanol/ sterile distilled water solution. Homogenize by agitating vigorously to obtain a blue concentrated solution. Refer to the PIF Supplement User Guide (Document #10000126749) for information on storage of diluted solution.

Sample Enrichment

Scope (Matrices)	Enrichment	Certification
Raw chicken breast, eggs, cantaloupe, and peanut butter	<i>n</i> g/ml of sample in 9 x <i>n</i> ml BPW Add RAPID' <i>Salmonella</i> Supplement 41.5 ± 1°C for 16–22 hr	AOAC
Broad range of food, animal feed, and environmental samples, 25 g	<i>n</i> g/ml of sample in 9 x <i>n</i> ml BPW Add RAPID' <i>Salmonella</i> Supplement 41.5 ± 1°C for 16–22 hr	NF VALIDATION
Infant formula and related dehydrated dairy ingredients, large sample size protocol, up to 375 g	<i>n</i> g/ml of sample in 9 x <i>n</i> ml prewarmed (37°C) BPW (without supplement) 37 ± 1°C for 20–24 hr	NF VALIDATION
Broad range of foods, including infant formula with and without probiotics, animal feed, and environmental samples	<i>n</i> g/ml of sample in 9 x <i>n</i> ml BPW Add RAPID' <i>Salmonella</i> Supplement 41.5 ± 1°C for 16–22 hr	NORDVAL
Dairy products, excluding raw milk	<i>n</i> g/ml of sample in 9 x <i>n</i> ml BPW (without supplement) 37 ± 1°C for 16–22 hr Transfer 0.5 ml to 10 ml of prewarmed RVS 41.5 ± 1°C for 6–26 hr	NF VALIDATION

Section 8 Confirmation of Positive Results

Infant formula and infant cereals with and without probiotics and ingredients, up to 375 g	<i>n</i> g of sample in 3x <i>n</i> ml pre-warmed (37°C) BPW Add PIF Supplement 37 ± 1°C for 18–24 hr	NF VALIDATION
Pet food and animal feed, up to 375 g	<i>n</i> g of sample in 5x <i>n</i> ml pre-warmed (37°C) BPW Add RAPID' <i>Salmonella</i> Supplement 37 ± 1°C for 18–24 hr	NF VALIDATION

1. Dilute *n* g or *n* ml of sample in required volume of BPW following the appropriate dilution ratio. See the table.
2. Homogenize with a stirrer/homogenizer.
3. Add the required volume of selective supplement. Selective supplement volume must be adapted to the volume of enrichment mix (sample + BPW).
4. Incubate according to Sample Enrichment table.
5. When following the EN ISO 6579-1 method, follow the sample enrichment protocol as written in the standard.
6. After the enrichment step, the broth can be stored at 2–8°C for 72 hr.

Inoculation and Plate Reading

1. Using a sterile loop, streak 10 µl of enrichment broth on RAPID' *Salmonella* Agar.
2. Incubate plates at 37 ± 1°C for 24 ± 2 hr.
3. *Salmonella* spp. form magenta colonies on RAPID' *Salmonella* Agar.
4. After the incubation step, plates can be stored at 2–8°C for 72 hr.

Section 8 Confirmation of Positive Results

1. In the context of AOAC validation, confirm suspect isolated colonies according to classic confirmation test procedure described in the standard reference methods.
2. In the context of NF VALIDATION and NORDVAL, all samples identified as positive must be confirmed in one of the following ways:
 - a. Using the conventional tests described in the CEN or ISO standard methods (including the purification step).
 - b. Using nucleic probes as described in the ISO 7218 standard (for example, iQ-Check *Salmonella* II Real-Time PCR Kit, catalog #3578123) using isolated colonies (with or without purification step).
 - c. Evaluating oxidase activity with an oxidase test followed by an Omnivalent Omni-O test (A–60) (*Salmonella* Omni-O Antiserum (A–60) catalog #3560781) using one to three isolated suspect colonies. If reaction to the Omni-O test is positive, proceed with an ONPG

Biochemical Test (catalog #3553822). *Salmonella* are negative to the oxidase test, positive to the Omni-O test (A–60), and negative to the ONPG Test, with the exception of lactose-positive *Salmonella*, which are ONPG positive.

- d. Performing a latex agglutination test using the *Salmonella* Latex Test (catalog #3556710) on an isolated colony or using the *Salmonella* Confirm Latex Test (catalog #3556711) on an isolated colony. Oxoid's *Salmonella* Latex Test has also been validated. *Salmonella* belonging to serogroups B to E and G are positive on the *Salmonella* Latex Test (catalog #3556710).
 - e. Using the MALDI Biotyper test (Bruker's Biotyper System includes a MALDI time-of-flight mass spectrometer and MBT Compass Software, version 4) directly from an isolated colony or after a purification step. The genus and/or species identification information given by the MALDI Biotyper Complete Solution is not included in the scope of the NF VALIDATION mark.
 - f. Using any other NF VALIDATION certified method based on a different principle from that of RAPID'*Salmonella* Agar. The validated protocol of the second method must be respected in its entirety. All steps preceding the detection step used as a starting point for confirmation must be common to both methods.
3. In the event of discordant results (presumptive positive with RAPID'*Salmonella*, negative with confirmation method, particularly with the latex test), the laboratory must follow the necessary steps to ensure validity of the results obtained.

Section 9

Confirmation of Other Methods

Confirmation of Positive Results with iQ-Check *Salmonella* II Kit (NF VALIDATION Certified Protocol)

1. Add 500 µl of enriched BPW to 10 ml of prewarmed RVS broth.
2. Incubate at $41.5 \pm 1^\circ\text{C}$ for 6–26 hr.
3. Using a sterile loop, streak 10 µl of enrichment broth on RAPID'*Salmonella* Agar.
4. Incubate plates at $37 \pm 1^\circ\text{C}$ for 24 ± 2 hr.
5. *Salmonella* spp. form magenta colonies on RAPID'*Salmonella* Agar. Perform a confirmation of one magenta colony using a latex agglutination test or using the conventional tests described in Section 8.2.

Section 10

Test Performance and Validations

Certification	Scope	Validation Protocol	Reference Protocol	Certificate Reference
AOAC-RI	Raw chicken breast, eggs, cantaloupe, and peanut butter	Performance Tested Methods	FDA BAM Chap. 5 USDA MLG 4.10	 License# 050701
NORDVAL	Broad range of foods, including infant formula with and without probiotics, animal feed, and environmental samples	ISO 16140-2:2016	EN ISO 6579:2002	 NORDVAL # 032
NF VALIDATION	Broad range of food, animal feed, and environmental samples (25 g) Infant formula and related dehydrated dairy ingredients, large sample size protocol, up to 375 g, 1:10 dilution ratio Infant formula and infant cereals with and without probiotics and ingredients, large sample size protocol (up to 375 g), 1:4 dilution ratio Pet food and animal feed, large sample size protocol (up to 375 g), 1:6 dilution ratio	EN ISO 16140-2	EN ISO 6579-1	 BRD: 07/11–12/05 ALTERNATIVE ANALYTICAL METHODS FOR AGRIBUSINESS Certified by AFNOR Certification http://nf-validation.afnor.org/en

Section 11 References

Cooke VM et al. (1999). A novel chromogenic ester agar medium for detection of *Salmonellae*. Appl Environ Microbiol, 65, 807812.

ISO 6579-1: 2017/ ADM 1: 2020 . Microbiology of the food chain — Horizontal method for the detection, enumeration and serotyping of *Salmonella* — Part 1: detection of *Salmonella* spp.

Manafi M et al. (1991). Fluorogenic and chromogenic substrates used in bacterial diagnostics. Microbiol Rev, 55, 335–348.

Perry JD et al. (1999). ABC medium, a new chromogenic agar for selective isolation of *Salmonella* spp. J Clin Microbiol, 37, 766–768.

United States Department of Agriculture. *Microbiology Laboratory Guidebook*. Chapter 4.10 Isolation and Identification of *Salmonella* from Meat, Poultry Pasteurized Egg, and Siluriformes (Fish) Products and Carcass and Environmental Sponges. <https://www.fsis.usda.gov/wps/wcm/connect/700c05fe-06a2-492a-a6e1-3357f7701f52/mlg-4.pdf?MOD=AJPERES>

United States Food and Drug Administration. Bacteriological Analytical Manual, Chapter 5 *Salmonella*. <http://www.fda.gov/Food/FoodScienceResearch/LaboratoryMethods/ucm070149.htm>

Section 12

Revision History

Release date	Document number	Change
October 2020	10000126748 Ver A	- Extension of validation for large sample size for PIF Supplement with 1:4 dilution. - New document design - Document number change – previous version RAPID <i>Salmonella</i> V11 25 June 2018
June 2021	10000126748 Ver B	New supplement format
January 2023	10000126748 Ver C	- Withdrawal of tablet supplement format - Selective supplement preparation update - Extension of NF Validation for pet food and animal feed up to 375 g

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10000126748 Ver C US/EG

RAPID'*Salmonella* Agar

User Guide

Chromogenic media for the detection of *Salmonella* spp. in the analysis of food products for human consumption, animal feed, and in environmental samples

Catalog #3563961, Prepared plates, 90 mm x 20 dishes
Catalog #3563963, Prepared plates, 90 mm x 120 dishes
Catalog #3564705, Dehydrated, 500 g



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Table of Contents

Section 1	Introduction	1
Section 2	RAPID' <i>Salmonella</i> Principle	1
Section 3	Theoretical Formula	1
Section 4	Shelf Life and Storage	1
Section 5	Materials Required but Not Supplied	1
	Equipment.....	2
	Supplies	2
Section 6	Precautions, Limitations of Use, and Quality Control	2
Section 7	Protocol.....	3
	Preparation of Dehydrated Medium.....	3
	Preparation of Selective Supplement	4
	Sample Enrichment	4
	Inoculation and Plate Reading.....	5
Section 8	Confirmation of Positive Results.....	5
Section 9	Confirmation of Other Methods.....	6
Section 10	Test Performance and Validations	6
Section 11	References.....	7
Section 12	Revision History.....	7

Section 1

Introduction

Salmonella is a gram-negative, non-spore-forming, rod-shaped bacterium that is predominantly motile, with only a few exceptions. *Salmonella* has been one of the two most frequently reported causes of bacterial foodborne illness. Common symptoms of salmonellosis include diarrhea, fever, and abdominal cramps occurring 12–72 hr after infection. Studies have shown that for each laboratory-confirmed case of *Salmonella*, 38 cases occur in the community that remain unreported. Testing for the presence of *Salmonella* in food and in the processing environment is an important practice. The use of chromogenic substrates in culture media increases the ease of use of the media with results based on color changing enzymatic reactions.

Section 2

RAPID'*Salmonella* Principle

RAPID'*Salmonella* Agar allows the presumptive identification of motile and non-motile *Salmonella* spp. by detecting C8-esterase activity. Simultaneous screening of β -glucosidase activity permits the differentiation of *Salmonella* colonies from those of other *Enterobacteriaceae*. After incubation, *Salmonella* appear as readily identifiable, typically magenta, colonies, whereas non-*Salmonella* bacteria grow as blue or colorless colonies. RAPID'*Salmonella* Agar permits detection of motile and non-motile *Salmonella*, as well as lactose-positive *Salmonella* Typhi and *Salmonella* Paratyphi.

Section 3

Theoretical Formula

Nutritive mix	14.5 g
Selective agents	14 g
Chromogenic mix	2.3 g
Agar	12.7 g
Distilled water	qsp 1,000 ml

Final pH at 25°C = 7.2 ± 0.2

Section 4

Shelf Life and Storage

- Dehydrated agar: 2–8°C in carefully sealed package and stored in a dry and dark place
- Prepoured agar: 2–8°C in a dark place
- Plate prepared from the dehydrated agar: 15 days at 2–8°C in carefully sealed package and stored in a dry and dark place

Section 5

Materials Required but Not Supplied

Equipment

- All usual laboratory equipment
- Hot plate
- Scale, sensitivity of 0.1 g
- Stirrer/homogenizer
- Thermostatically controlled incubator or incubation room, precise to $\pm 1^{\circ}\text{C}$
- Water bath

Supplies

- Enrichment medium: Buffered Peptone Water (BPW) (catalog #3554179, 225 ml x 6 bottles; 3564684, dehydrated 500 g; 3555790, 5 L x 2 bags; 3555795, 3 L x 4 bags)
- Inoculating loops
- PIF Supplement (catalog #12013322, 2 g bottle)
- RAPID'*Salmonella* selective supplement for short protocol (catalog #3564709, 10x concentrated x 100 capsules; 3564710, 1x concentrated x 100 capsules; 3564712, 1 box x 100 tests)
- Rappaport Vassiliadis Soy (RVS) for double enrichment protocol (catalog #3555773, 10 ml x 25 tubes x 25; 3564324, dehydrated 500 g)
- Sterile petri dishes
- Sterile pipets
- Sterile weigh bags

Section 6

Precautions, Limitations of Use, and Quality Control

Precautions

- Respect Good Laboratory Practice (EN ISO 7218). Appropriate protection, such as gloves and lab coats, should be worn when working with potentially infectious live bacteria such as *Salmonella*
- Media that have come in contact with food samples should be considered contaminated and should be disposed of in accordance with local rules and regulations

- If the whole selective supplement capsule is added to the BPW, sterile tweezers must be used to add the capsule to the bag. (We recommend that you check that the capsule actually opened during the stomaching stage.) If the capsule contents are handled with the fingers, it cannot be added to the enrichment broth due to the risk of contamination
- The RAPID'*Salmonella* Capsules and RAPID'*Salmonella* Supplement Tablets contain selective agents and an excipient. Selective agents dissolve very well. The excipient, however, remains in suspension and may create a deposit when the contents of the capsule are diluted in a small quantity of BPW or sterile distilled water. Always shake well, therefore, before using the concentrated solution

Limitations of Use

- ONPG confirmation test excludes confirmation of lactose-positive *Salmonella*
- Although the most prevalent *Salmonella* strains can be detected by *Salmonella* Confirm Latex Kit, the kit did not detect 41 of the 150 strains tested during the NF VALIDATION extension of RAPID'*Salmonella*
- Some strains of *Salmonella* (a few are part of *Salmonella* serovar Dublin and *S. bongori* species) can show a weak magenta color due to low esterase activity
- In the context of NORDVAL validation, the double enrichment protocol was validated with an RVS enrichment time of 24 ± 2 hr only
- After the incubation step, the plate can be stored at 2–8°C for 72 hr. Some non-*Salmonella* strains (for example, *E. coli*) can develop pink-colored colonies after this storage. These doubtful colonies give a negative confirmation test

Quality Control

- Every product manufactured and marketed by Bio-Rad is subject to a quality assurance procedure at all stages, from reception of raw materials through to marketing of the finished products. Each batch of finished product undergoes quality control according to EN ISO 11133 and is marketed only if it satisfies the acceptability criteria. Documentation relative to the production and quality control of each batch is kept on file
- For SDS product safety information and certificate of analysis, visit www.bio-rad.com

Section 7 Protocol

Preparation of Dehydrated Medium

1. Always shake bottle before use.
2. Dissolve 43.5 g of powder in 1 L of distilled water and mix until a homogenous suspension is obtained.
3. Heat gently, agitating frequently, then bring to a boil for less than 1 min.

4. Do not prolong heating. **Do not autoclave.**
5. Cool the medium to 50°C. Dispense in petri dishes and leave on the bench to dry.
6. One 500 g bottle of powder makes 11.5 L of medium.

Preparation of Selective Supplement

1. One capsule (3564710) is ready to use for 25 g or 25 ml of sample diluted in 225 ml of BPW. Either the whole capsule or its contents can be added to the sample bag before the mixing step. In order to make handling easier, we recommend opening the capsule and pouring out the contents.
2. Concentrated (10x) capsules (3564709) must be diluted before use. Pour the contents of one capsule into 10 ml of BPW or sterile distilled water. Homogenize to obtain a red concentrated solution. Use 1 ml of concentrated solution for 25 g or 25 ml of sample diluted in 225 ml BPW. For large sample size, adjust the selective supplement volume to the volume of enrichment mix to obtain concentration of 4:1000 g/ml. Diluted solution can be stored for 1 week at ambient temperature or at 2–8°C.
3. Boxed supplement (3564712) is concentrated and must be diluted before use. Open the box and fill with 100 ml of BPW or sterile distilled water. Homogenize by agitating vigorously in order to obtain a red concentrated solution. Use 1 ml of concentrated solution for 25 g or 25 ml of sample diluted in 225 ml BPW. For large sample size, adjust the selective supplement volume to the volume of enrichment mix to obtain concentration of 4:1000 g/ml. Diluted solution can be stored for 1 week at ambient temperature or at 2–8°C.
4. Open the bottle of PIF Supplement (12013322) and fill with 40 ml of a 1:1 ethanol/ sterile distilled water solution. Homogenize by agitating vigorously to obtain a blue concentrated solution. Refer to the PIF Supplement User Guide (Document #10000126749) for information on storage of diluted solution.

Sample Enrichment

Scope (Matrices)	Enrichment	Certification
Raw chicken breast, eggs, cantaloupe, and peanut butter	<i>n</i> g/ml of sample in 9 x <i>n</i> ml BPW Add RAPID' <i>Salmonella</i> Supplement 41.5 ± 1°C for 16–22 hr	AOAC
Broad range of food, animal feed, and environmental samples, 25 g	<i>n</i> g/ml of sample in 9 x <i>n</i> ml BPW Add RAPID' <i>Salmonella</i> Supplement 41.5 ± 1°C for 16–22 hr	NF VALIDATION
Infant formula and related dehydrated dairy ingredients, large sample size protocol, up to 375 g	<i>n</i> g/ml of sample in 9 x <i>n</i> ml prewarmed (37°C) BPW (without supplement) 37 ± 1°C for 20–24 hr	NF VALIDATION
Broad range of foods, including infant formula with and without probiotics, animal feed, and environmental samples	<i>n</i> g/ml of sample in 9 x <i>n</i> ml BPW Add RAPID' <i>Salmonella</i> Supplement 41.5 ± 1°C for 16–22 hr	NORDVAL
Dairy products, excluding raw milk	<i>n</i> g/ml of sample in 9 x <i>n</i> ml BPW (without supplement) 37 ± 1°C for 16–22 hr Transfer 0.5 ml to 10 ml of prewarmed RVS 41.5 ± 1°C for 6–26 hr	NF VALIDATION

Section 8 Confirmation of Positive Results

Infant formula and infant cereals with and without probiotics and ingredients, up to 375 g	<i>n</i> g of sample in 3x <i>n</i> ml pre-warmed (37°C) BPW Add PIF Supplement 37 ± 1°C for 18–24 hr	NF VALIDATION
Pet food and animal feed, up to 375 g	<i>n</i> g of sample in 5x <i>n</i> ml pre-warmed (37°C) BPW Add RAPID' <i>Salmonella</i> Supplement 37 ± 1°C for 18–24 hr	NF VALIDATION

1. Dilute *n* g or *n* ml of sample in required volume of BPW following the appropriate dilution ratio. See the table.
2. Homogenize with a stirrer/homogenizer.
3. Add the required volume of selective supplement. Selective supplement volume must be adapted to the volume of enrichment mix (sample + BPW).
4. Incubate according to Sample Enrichment table.
5. When following the EN ISO 6579-1 method, follow the sample enrichment protocol as written in the standard.
6. After the enrichment step, the broth can be stored at 2–8°C for 72 hr.

Inoculation and Plate Reading

1. Using a sterile loop, streak 10 µl of enrichment broth on RAPID' *Salmonella* Agar.
2. Incubate plates at 37 ± 1°C for 24 ± 2 hr.
3. *Salmonella* spp. form magenta colonies on RAPID' *Salmonella* Agar.
4. After the incubation step, plates can be stored at 2–8°C for 72 hr.

Section 8 Confirmation of Positive Results

1. In the context of AOAC validation, confirm suspect isolated colonies according to classic confirmation test procedure described in the standard reference methods.
2. In the context of NF VALIDATION and NORDVAL, all samples identified as positive must be confirmed in one of the following ways:
 - a. Using the conventional tests described in the CEN or ISO standard methods (including the purification step).
 - b. Using nucleic probes as described in the ISO 7218 standard (for example, iQ-Check *Salmonella* II Real-Time PCR Kit, catalog #3578123) using isolated colonies (with or without purification step).
 - c. Evaluating oxidase activity with an oxidase test followed by an Omnivalent Omni-O test (A–60) (*Salmonella* Omni-O Antiserum (A–60) catalog #3560781) using one to three isolated suspect colonies. If reaction to the Omni-O test is positive, proceed with an ONPG

Biochemical Test (catalog #3553822). *Salmonella* are negative to the oxidase test, positive to the Omni-O test (A–60), and negative to the ONPG Test, with the exception of lactose-positive *Salmonella*, which are ONPG positive.

- d. Performing a latex agglutination test using the *Salmonella* Latex Test (catalog #3556710) on an isolated colony or using the *Salmonella* Confirm Latex Test (catalog #3556711) on an isolated colony. Oxoid's *Salmonella* Latex Test has also been validated. *Salmonella* belonging to serogroups B to E and G are positive on the *Salmonella* Latex Test (catalog #3556710).
 - e. Using the MALDI Biotyper test (Bruker's Biotyper System includes a MALDI time-of-flight mass spectrometer and MBT Compass Software, version 4) directly from an isolated colony or after a purification step. The genus and/or species identification information given by the MALDI Biotyper Complete Solution is not included in the scope of the NF VALIDATION mark.
 - f. Using any other NF VALIDATION certified method based on a different principle from that of RAPID'*Salmonella* Agar. The validated protocol of the second method must be respected in its entirety. All steps preceding the detection step used as a starting point for confirmation must be common to both methods.
3. In the event of discordant results (presumptive positive with RAPID'*Salmonella*, negative with confirmation method, particularly with the latex test), the laboratory must follow the necessary steps to ensure validity of the results obtained.

Section 9

Confirmation of Other Methods

Confirmation of Positive Results with iQ-Check *Salmonella* II Kit (NF VALIDATION Certified Protocol)

1. Add 500 µl of enriched BPW to 10 ml of prewarmed RVS broth.
2. Incubate at $41.5 \pm 1^\circ\text{C}$ for 6–26 hr.
3. Using a sterile loop, streak 10 µl of enrichment broth on RAPID'*Salmonella* Agar.
4. Incubate plates at $37 \pm 1^\circ\text{C}$ for 24 ± 2 hr.
5. *Salmonella* spp. form magenta colonies on RAPID'*Salmonella* Agar. Perform a confirmation of one magenta colony using a latex agglutination test or using the conventional tests described in Section 8.2.

Section 10

Test Performance and Validations

Section 11 References

Certification	Scope	Validation Protocol	Reference Protocol	Certificate Reference
AOAC-RI	Raw chicken breast, eggs, cantaloupe, and peanut butter	Performance Tested Methods	FDA BAM Chap. 5 USDA MLG 4.10	 License# 050701
NORDVAL	Broad range of foods, including infant formula with and without probiotics, animal feed, and environmental samples	ISO 16140-2:2016	EN ISO 6579:2002	 NORDVAL # 032
NF VALIDATION	Broad range of food, animal feed, and environmental samples (25 g) Infant formula and related dehydrated dairy ingredients, large sample size protocol, up to 375 g, 1:10 dilution ratio Infant formula and infant cereals with and without probiotics and ingredients, large sample size protocol (up to 375 g), 1:4 dilution ratio Pet food and animal feed, large sample size protocol (up to 375 g), 1:6 dilution ratio	EN ISO 16140-2	EN ISO 6579-1	 BRD: 07/11–12/05 ALTERNATIVE ANALYTICAL METHODS FOR AGRIBUSINESS Certified by AFNOR Certification http://nf-validation.afnor.org/en

Section 11 References

Cooke VM et al. (1999). A novel chromogenic ester agar medium for detection of *Salmonellae*. Appl Environ Microbiol, 65, 807812.

ISO 6579-1: 2017/ ADM 1: 2020 . Microbiology of the food chain — Horizontal method for the detection, enumeration and serotyping of *Salmonella* — Part 1: detection of *Salmonella* spp.

Manafi M et al. (1991). Fluorogenic and chromogenic substrates used in bacterial diagnostics. Microbiol Rev, 55, 335–348.

Perry JD et al. (1999). ABC medium, a new chromogenic agar for selective isolation of *Salmonella* spp. J Clin Microbiol, 37, 766–768.

United States Department of Agriculture. *Microbiology Laboratory Guidebook*. Chapter 4.10 Isolation and Identification of *Salmonella* from Meat, Poultry Pasteurized Egg, and Siluriformes (Fish) Products and Carcass and Environmental Sponges. <https://www.fsis.usda.gov/wps/wcm/connect/700c05fe-06a2-492a-a6e1-3357f7701f52/mlg-4.pdf?MOD=AJPERES>

United States Food and Drug Administration. Bacteriological Analytical Manual, Chapter 5 *Salmonella*. <http://www.fda.gov/Food/FoodScienceResearch/LaboratoryMethods/ucm070149.htm>

Section 12

Revision History

Release date	Document number	Change
October 2020	10000126748 Ver A	- Extension of validation for large sample size for PIF Supplement with 1:4 dilution. - New document design - Document number change – previous version RAPID <i>Salmonella</i> V11 25 June 2018
June 2021	10000126748 Ver B	New supplement format
January 2023	10000126748 Ver C	- Withdrawal of tablet supplement format - Selective supplement preparation update - Extension of NF Validation for pet food and animal feed up to 375 g

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10000126748 Ver C US/EG

RAPID'*Salmonella* Agar

User Guide

Chromogenic media for the detection of *Salmonella* spp. in the analysis of food products for human consumption, animal feed, and in environmental samples

Catalog #3563961, Prepared plates, 90 mm x 20 dishes
Catalog #3563963, Prepared plates, 90 mm x 120 dishes
Catalog #3564705, Dehydrated, 500 g



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Table of Contents

Section 1	Introduction	1
Section 2	RAPID' <i>Salmonella</i> Principle	1
Section 3	Theoretical Formula.....	1
Section 4	Shelf Life and Storage	1
Section 5	Materials Required but Not Supplied	1
	Equipment.....	2
	Supplies	2
Section 6	Precautions, Limitations of Use, and Quality Control	2
Section 7	Protocol.....	3
	Preparation of Dehydrated Medium.....	3
	Preparation of Selective Supplement	4
	Sample Enrichment	4
	Inoculation and Plate Reading.....	5
Section 8	Confirmation of Positive Results.....	5
Section 9	Confirmation of Other Methods.....	6
Section 10	Test Performance and Validations	6
Section 11	References.....	7
Section 12	Revision History.....	7

Section 1

Introduction

Salmonella is a gram-negative, non-spore-forming, rod-shaped bacterium that is predominantly motile, with only a few exceptions. *Salmonella* has been one of the two most frequently reported causes of bacterial foodborne illness. Common symptoms of salmonellosis include diarrhea, fever, and abdominal cramps occurring 12–72 hr after infection. Studies have shown that for each laboratory-confirmed case of *Salmonella*, 38 cases occur in the community that remain unreported. Testing for the presence of *Salmonella* in food and in the processing environment is an important practice. The use of chromogenic substrates in culture media increases the ease of use of the media with results based on color changing enzymatic reactions.

Section 2

RAPID'*Salmonella* Principle

RAPID'*Salmonella* Agar allows the presumptive identification of motile and non-motile *Salmonella* spp. by detecting C8-esterase activity. Simultaneous screening of β -glucosidase activity permits the differentiation of *Salmonella* colonies from those of other *Enterobacteriaceae*. After incubation, *Salmonella* appear as readily identifiable, typically magenta, colonies, whereas non-*Salmonella* bacteria grow as blue or colorless colonies. RAPID'*Salmonella* Agar permits detection of motile and non-motile *Salmonella*, as well as lactose-positive *Salmonella* Typhi and *Salmonella* Paratyphi.

Section 3

Theoretical Formula

Nutritive mix	14.5 g
Selective agents	14 g
Chromogenic mix	2.3 g
Agar	12.7 g
Distilled water	qsp 1,000 ml

Final pH at 25°C = 7.2 $\pm$ 0.2

Section 4

Shelf Life and Storage

- Dehydrated agar: 2–8°C in carefully sealed package and stored in a dry and dark place
- Prepoured agar: 2–8°C in a dark place
- Plate prepared from the dehydrated agar: 15 days at 2–8°C in carefully sealed package and stored in a dry and dark place

Section 5

Materials Required but Not Supplied

Equipment

- All usual laboratory equipment
- Hot plate
- Scale, sensitivity of 0.1 g
- Stirrer/homogenizer
- Thermostatically controlled incubator or incubation room, precise to $\pm 1^{\circ}\text{C}$
- Water bath

Supplies

- Enrichment medium: Buffered Peptone Water (BPW) (catalog #3554179, 225 ml x 6 bottles; 3564684, dehydrated 500 g; 3555790, 5 L x 2 bags; 3555795, 3 L x 4 bags)
- Inoculating loops
- PIF Supplement (catalog #12013322, 2 g bottle)
- RAPID'*Salmonella* selective supplement for short protocol (catalog #3564709, 10x concentrated x 100 capsules; 3564710, 1x concentrated x 100 capsules; 3564712, 1 box x 100 tests)
- Rappaport Vassiliadis Soy (RVS) for double enrichment protocol (catalog #3555773, 10 ml x 25 tubes x 25; 3564324, dehydrated 500 g)
- Sterile petri dishes
- Sterile pipets
- Sterile weigh bags

Section 6

Precautions, Limitations of Use, and Quality Control

Precautions

- Respect Good Laboratory Practice (EN ISO 7218). Appropriate protection, such as gloves and lab coats, should be worn when working with potentially infectious live bacteria such as *Salmonella*
- Media that have come in contact with food samples should be considered contaminated and should be disposed of in accordance with local rules and regulations

- If the whole selective supplement capsule is added to the BPW, sterile tweezers must be used to add the capsule to the bag. (We recommend that you check that the capsule actually opened during the stomaching stage.) If the capsule contents are handled with the fingers, it cannot be added to the enrichment broth due to the risk of contamination
- The RAPID'*Salmonella* Capsules and RAPID'*Salmonella* Supplement Tablets contain selective agents and an excipient. Selective agents dissolve very well. The excipient, however, remains in suspension and may create a deposit when the contents of the capsule are diluted in a small quantity of BPW or sterile distilled water. Always shake well, therefore, before using the concentrated solution

Limitations of Use

- ONPG confirmation test excludes confirmation of lactose-positive *Salmonella*
- Although the most prevalent *Salmonella* strains can be detected by *Salmonella* Confirm Latex Kit, the kit did not detect 41 of the 150 strains tested during the NF VALIDATION extension of RAPID'*Salmonella*
- Some strains of *Salmonella* (a few are part of *Salmonella* serovar Dublin and *S. bongori* species) can show a weak magenta color due to low esterase activity
- In the context of NORDVAL validation, the double enrichment protocol was validated with an RVS enrichment time of 24 ± 2 hr only
- After the incubation step, the plate can be stored at 2–8°C for 72 hr. Some non-*Salmonella* strains (for example, *E. coli*) can develop pink-colored colonies after this storage. These doubtful colonies give a negative confirmation test

Quality Control

- Every product manufactured and marketed by Bio-Rad is subject to a quality assurance procedure at all stages, from reception of raw materials through to marketing of the finished products. Each batch of finished product undergoes quality control according to EN ISO 11133 and is marketed only if it satisfies the acceptability criteria. Documentation relative to the production and quality control of each batch is kept on file
- For SDS product safety information and certificate of analysis, visit www.bio-rad.com

Section 7 Protocol

Preparation of Dehydrated Medium

1. Always shake bottle before use.
2. Dissolve 43.5 g of powder in 1 L of distilled water and mix until a homogenous suspension is obtained.
3. Heat gently, agitating frequently, then bring to a boil for less than 1 min.

4. Do not prolong heating. **Do not autoclave.**
5. Cool the medium to 50°C. Dispense in petri dishes and leave on the bench to dry.
6. One 500 g bottle of powder makes 11.5 L of medium.

Preparation of Selective Supplement

1. One capsule (3564710) is ready to use for 25 g or 25 ml of sample diluted in 225 ml of BPW. Either the whole capsule or its contents can be added to the sample bag before the mixing step. In order to make handling easier, we recommend opening the capsule and pouring out the contents.
2. Concentrated (10x) capsules (3564709) must be diluted before use. Pour the contents of one capsule into 10 ml of BPW or sterile distilled water. Homogenize to obtain a red concentrated solution. Use 1 ml of concentrated solution for 25 g or 25 ml of sample diluted in 225 ml BPW. For large sample size, adjust the selective supplement volume to the volume of enrichment mix to obtain concentration of 4:1000 g/ml. Diluted solution can be stored for 1 week at ambient temperature or at 2–8°C.
3. Boxed supplement (3564712) is concentrated and must be diluted before use. Open the box and fill with 100 ml of BPW or sterile distilled water. Homogenize by agitating vigorously in order to obtain a red concentrated solution. Use 1 ml of concentrated solution for 25 g or 25 ml of sample diluted in 225 ml BPW. For large sample size, adjust the selective supplement volume to the volume of enrichment mix to obtain concentration of 4:1000 g/ml. Diluted solution can be stored for 1 week at ambient temperature or at 2–8°C.
4. Open the bottle of PIF Supplement (12013322) and fill with 40 ml of a 1:1 ethanol/ sterile distilled water solution. Homogenize by agitating vigorously to obtain a blue concentrated solution. Refer to the PIF Supplement User Guide (Document #10000126749) for information on storage of diluted solution.

Sample Enrichment

Scope (Matrices)	Enrichment	Certification
Raw chicken breast, eggs, cantaloupe, and peanut butter	n g/ml of sample in 9 x n ml BPW Add RAPID' <i>Salmonella</i> Supplement 41.5 ± 1°C for 16–22 hr	AOAC
Broad range of food, animal feed, and environmental samples, 25 g	n g/ml of sample in 9 x n ml BPW Add RAPID' <i>Salmonella</i> Supplement 41.5 ± 1°C for 16–22 hr	NF VALIDATION
Infant formula and related dehydrated dairy ingredients, large sample size protocol, up to 375 g	n g/ml of sample in 9 x n ml prewarmed (37°C) BPW (without supplement) 37 ± 1°C for 20–24 hr	NF VALIDATION
Broad range of foods, including infant formula with and without probiotics, animal feed, and environmental samples	n g/ml of sample in 9 x n ml BPW Add RAPID' <i>Salmonella</i> Supplement 41.5 ± 1°C for 16–22 hr	NORDVAL
Dairy products, excluding raw milk	n g/ml of sample in 9 x n ml BPW (without supplement) 37 ± 1°C for 16–22 hr Transfer 0.5 ml to 10 ml of prewarmed RVS 41.5 ± 1°C for 6–26 hr	NF VALIDATION

Section 8 Confirmation of Positive Results

Infant formula and infant cereals with and without probiotics and ingredients, up to 375 g	<i>n</i> g of sample in 3x <i>n</i> ml pre-warmed (37°C) BPW Add PIF Supplement 37 ± 1°C for 18–24 hr	NF VALIDATION
Pet food and animal feed, up to 375 g	<i>n</i> g of sample in 5x <i>n</i> ml pre-warmed (37°C) BPW Add RAPID' <i>Salmonella</i> Supplement 37 ± 1°C for 18–24 hr	NF VALIDATION

1. Dilute *n* g or *n* ml of sample in required volume of BPW following the appropriate dilution ratio. See the table.
2. Homogenize with a stirrer/homogenizer.
3. Add the required volume of selective supplement. Selective supplement volume must be adapted to the volume of enrichment mix (sample + BPW).
- 4.
5. Incubate according to Sample Enrichment table.
6. When following the EN ISO 6579-1 method, follow the sample enrichment protocol as written in the standard.
7. After the enrichment step, the broth can be stored at 2–8°C for 72 hr.

Inoculation and Plate Reading

1. Using a sterile loop, streak 10 µl of enrichment broth on RAPID'*Salmonella* Agar.
2. Incubate plates at 37 ± 1°C for 24 ± 2 hr.
3. *Salmonella* spp. form magenta colonies on RAPID'*Salmonella* Agar.
4. After the incubation step, plates can be stored at 2–8°C for 72 hr.

Section 8 Confirmation of Positive Results

1. In the context of AOAC validation, confirm suspect isolated colonies according to classic confirmation test procedure described in the standard reference methods.
2. In the context of NF VALIDATION and NORDVAL, all samples identified as positive must be confirmed in one of the following ways:
 - a. Using the conventional tests described in the CEN or ISO standard methods (including the purification step).
 - b. Using nucleic probes as described in the ISO 7218 standard (for example, iQ-Check *Salmonella* II Real-Time PCR Kit, catalog #3578123) using isolated colonies (with or without purification step).
 - c. Evaluating oxidase activity with an oxidase test followed by an Omnivalent Omni-O test (A–60) (*Salmonella* Omni-O Antiserum (A–60) catalog #3560781) using one to three isolated

suspect colonies. If reaction to the Omni-O test is positive, proceed with an ONPG Biochemical Test (catalog #3553822). *Salmonella* are negative to the oxidase test, positive to the Omni-O test (A–60), and negative to the ONPG Test, with the exception of lactose-positive *Salmonella*, which are ONPG positive.

- d. Performing a latex agglutination test using the *Salmonella* Latex Test (catalog #3556710) on an isolated colony or using the *Salmonella* Confirm Latex Test (catalog #3556711) on an isolated colony. Oxoid's *Salmonella* Latex Test has also been validated. *Salmonella* belonging to serogroups B to E and G are positive on the *Salmonella* Latex Test (catalog #3556710).
 - e. Using the MALDI Biotyper test (Bruker's Biotyper System includes a MALDI time-of-flight mass spectrometer and MBT Compass Software, version 4) directly from an isolated colony or after a purification step. The genus and/or species identification information given by the MALDI Biotyper Complete Solution is not included in the scope of the NF VALIDATION mark.
 - f. Using any other NF VALIDATION certified method based on a different principle from that of RAPID'*Salmonella* Agar. The validated protocol of the second method must be respected in its entirety. All steps preceding the detection step used as a starting point for confirmation must be common to both methods.
3. In the event of discordant results (presumptive positive with RAPID'*Salmonella*, negative with confirmation method, particularly with the latex test), the laboratory must follow the necessary steps to ensure validity of the results obtained.

Section 9

Confirmation of Other Methods

Confirmation of Positive Results with iQ-Check *Salmonella* II Kit (NF VALIDATION Certified Protocol)

1. Add 500 µl of enriched BPW to 10 ml of prewarmed RVS broth.
2. Incubate at $41.5 \pm 1^\circ\text{C}$ for 6–26 hr.
3. Using a sterile loop, streak 10 µl of enrichment broth on RAPID'*Salmonella* Agar.
4. Incubate plates at $37 \pm 1^\circ\text{C}$ for 24 ± 2 hr.
5. *Salmonella* spp. form magenta colonies on RAPID'*Salmonella* Agar. Perform a confirmation of one magenta colony using a latex agglutination test or using the conventional tests described in Section 8.2.

Section 10

Test Performance and Validations

Certification	Scope	Validation Protocol	Reference Protocol	Certificate Reference
AOAC-RI	Raw chicken breast, eggs, cantaloupe, and peanut butter	Performance Tested Methods	FDA BAM Chap. 5 USDA MLG 4.10	 License# 050701
NORDVAL	Broad range of foods, including infant formula with and without probiotics, animal feed, and environmental samples	ISO 16140-2:2016	EN ISO 6579:2002	 NORDVAL # 032
NF VALIDATION	Broad range of food, animal feed, and environmental samples (25 g) Infant formula and related dehydrated dairy ingredients, large sample size protocol, up to 375 g, 1:10 dilution ratio Infant formula and infant cereals with and without probiotics and ingredients, large sample size protocol (up to 375 g), 1:4 dilution ratio Pet food and animal feed, large sample size protocol (up to 375 g), 1:6 dilution ratio	EN ISO 16140-2	EN ISO 6579-1	 BRD: 07/11–12/05 ALTERNATIVE ANALYTICAL METHODS FOR AGRIBUSINESS Certified by AFNOR Certification http://nf-validation.afnor.org/en

Section 11

References

Cooke VM et al. (1999). A novel chromogenic ester agar medium for detection of *Salmonellae*. Appl Environ Microbiol, 65, 807812.

ISO 6579-1: 2017/ ADM 1: 2020 . Microbiology of the food chain — Horizontal method for the detection, enumeration and serotyping of *Salmonella* — Part 1: detection of *Salmonella* spp.

Manafi M et al. (1991). Fluorogenic and chromogenic substrates used in bacterial diagnostics. Microbiol Rev, 55, 335–348.

Perry JD et al. (1999). ABC medium, a new chromogenic agar for selective isolation of *Salmonella* spp. J Clin Microbiol, 37, 766–768.

United States Department of Agriculture. *Microbiology Laboratory Guidebook*. Chapter 4.10 Isolation and Identification of *Salmonella* from Meat, Poultry Pasteurized Egg, and Siluriformes (Fish) Products and Carcass and Environmental Sponges. <https://www.fsis.usda.gov/wps/wcm/connect/700c05fe-06a2-492a-a6e1-3357f7701f52/mlg-4.pdf?MOD=AJPERES>

United States Food and Drug Administration. Bacteriological Analytical Manual, Chapter 5 *Salmonella*. <http://www.fda.gov/Food/FoodScienceResearch/LaboratoryMethods/ucm070149.htm>

Section 12

Revision History

Release date	Document number	Change
October 2020	10000126748 Ver A	- Extension of validation for large sample size for PIF Supplement with 1:4 dilution. - New document design - Document number change – previous version RAPID_Salmonella_V11_25 June 2018
June 2021	10000126748 Ver B	New supplement format
January 2023	10000126748 Ver C	- Withdrawal of tablet supplement format - Selective supplement preparation update - Extension of NF Validation for pet food and animal feed up to 375 g

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New Zealand 64 9 415 2280 Norway 00 800 00 24 67 23 Poland 00 800 00 24 67 23 Portugal 00 800 00 24 67 23
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United Arab Emirates 36 1 459 6150 United Kingdom 00 800 00 24 67 23

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RAPID'*Salmonella* Agar

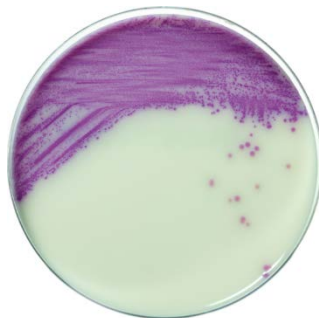
Guide d'utilisation

Milieu chromogénique pour la détection des *Salmonella* spp. dans l'analyse des produits alimentaires destinés à la consommation humaine ou animale et des échantillons environnementaux

N° de référence 3563961, boîte préparée, 90 mm x 20 boîtes

N° de référence 3563963, boîte préparée, 90 mm x 120 boîtes

N° de référence 3564705, base déshydratée, 500 g



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Sommaire

Section 1	Introduction	1
Section 2	RAPID' <i>Salmonella</i> - Principe	1
Section 3	Formule théorique	1
Section 4	Durée de conservation et stockage	1
Section 5	Matériel requis non fourni	2
	Matériel	2
	Produits.....	2
Section 6	Précautions, limites d'utilisation et contrôle qualité	3
Section 7	Protocole	4
	Préparation du milieu de culture déshydraté	4
	Préparation du supplément sélectif	4
	Enrichissement de l'échantillon.....	5
	Inoculation et lecture	6
Section 8	Confirmation des résultats positifs	7
Section 9	Confirmation d'autres méthodes	8
Section 10	Performance du test et validations	8
Section 11	Références	9
Section 12	Historique des révisions	10

Section 1 Introduction

Les *Salmonelles* sont des bacilles à Gram négatif, non sporulantes et majoritairement mobiles (à quelques exceptions près). Elles constituent l'une des deux causes les plus fréquentes de maladies d'origine alimentaire. Les symptômes courants de la salmonellose sont des diarrhées, une fièvre et des douleurs abdominales survenant dans les 12 à 72 heures après l'infection. Les études ont montré que pour chaque cas de *Salmonelle* confirmé en laboratoire, 38 cas apparaissent dans la communauté sans être rapportés. La détection des *Salmonelles* dans les aliments et l'environnement de transformation alimentaire est primordiale. L'utilisation de substrats chromogènes dans le milieu de culture augmente la simplicité de lecture grâce aux résultats basés sur des réactions enzymatiques avec changement de couleur.

Section 2 RAPID'*Salmonella* - Principe

RAPID'*Salmonella* Agar permet l'identification présumée des *Salmonella* spp. mobiles et non mobiles grâce à la détection de l'activité de la C8-estérase. La détection simultanée de l'activité de la β -glucosidase permet de distinguer les colonies de *Salmonelle* des autres colonies d'*Enterobacteriaceae*. Après incubation, les salmonelles apparaissent sous forme de colonies de couleur magenta typique, facilement identifiables, alors que les autres colonies sont bleues ou incolores. RAPID'*Salmonella* Agar permet de détecter les *Salmonella* mobiles et non mobiles, ainsi que les *Salmonella* lactose positives, *Salmonella* Typhi et *Salmonella* Paratyphi.

Section 3 Formule théorique

Mélange nutritif	14,5 g
Agents sélectifs	14 g
Mélange chromogène	2,3 g
Agar	12,7 g
Eau distillée	q.s.p. 1 000 ml

pH final à 25 °C = 7,2 $\pm$ 0,2

Section 4 Durée de conservation et stockage

- Gélose base déshydratée : 2–8 °C en emballage soigneusement scellé, dans un endroit sec et à l'abri de la lumière
- Gélose précoulée : 2–8 °C à l'abri de la lumière
- Boîte préparée à partir de la gélose base déshydratée : 15 jours à 2–8 °C, en emballage soigneusement scellé, dans un endroit sec et à l'abri de la lumière

Section 5

Matériel requis non fourni

Matériel

- Tout le matériel de laboratoire habituel
- Plaque chauffante
- Balance, sensibilité 0,1 g
- Agitateur/homogénéisateur
- Étuve ou enceinte thermostatique, précision ± 1 °C
- Bain-marie

Produits

- Milieu d'enrichissement : Eau peptonée tamponnée (EPT) (n° de référence 3554179, 225 ml x 6 flacons ; 3564684, base déshydratée 500 g ; 3555790, 5 L x 2 poches ; 3555795, 3 L x 4 poches)
- Anses d'inoculation
- PIF Supplement (n° de référence 12013322, 2 g)
- Supplément sélectif RAPID'*Salmonella* pour protocole court (n° de référence 3564709, concentration 10 x 100 capsules ; 3564710, concentration 1 x 100 capsules ; 3564712, 1 boîte x 100 tests)
- Milieu Rappaport Vassiliadis Soy (RVS) pour le protocole de doubleenrichissement (n° de référence 3555773, 10 ml x 25 tubes ; 3564324, base déshydratée 500 g)
- Boîtes de Pétri stériles
- Pipettes stériles
- Sacs de pesée stériles

Section 6

Précautions, limites d'utilisation et contrôle qualité

Précautions

- Respecter les bonnes pratiques de laboratoire (EN ISO 7218). Porter un équipement de protection approprié, par exemple des gants et une blouse de laboratoire, pour travailler avec des bactéries vivantes potentiellement infectieuses telles que les *Salmonella*.
- Les milieux qui sont entrés en contact avec des échantillons alimentaires doivent être considérés comme contaminés et doivent être éliminés conformément aux règles et réglementations locales.
- Si la capsule entière de supplément sélectif est ajoutée à l'eau peptonée tamponnée, il est nécessaire d'utiliser des pincettes stériles pour ajouter la capsule à la poche. (Nous recommandons de vérifier l'ouverture effective de la capsule pendant l'étape en Stomacher.) Si le contenu de la capsule est manipulé avec les doigts, il ne peut pas être ajouté au bouillon d'enrichissement en raison du risque de contamination.
- Les capsules RAPID'*Salmonella* et les comprimés RAPID'*Salmonella* contiennent des agents sélectifs et un excipient. Les agents sélectifs se dissolvent très bien. L'excipient, cependant, reste en suspension et peut créer un dépôt lorsque le contenu de la capsule est dilué dans une petite quantité d'eau peptonée tamponnée ou d'eau distillée stérile. Aussi, il est nécessaire de bien agiter avant d'utiliser la solution concentrée.

Limites d'utilisation

- Le test de confirmation ONPG exclut la confirmation des *Salmonella* lactose positives.
- Bien que les souches de *Salmonella* les plus courantes puissent être détectées par le kit *Salmonella* Confirm Latex, celui-ci n'a pas détecté 41 des 150 souches testées au cours de l'extension NF VALIDATION de RAPID'*Salmonella*.
- Certaines souches de *Salmonella* (quelques-unes font partie des espèces *Salmonella* sérotype Dublin et *S. bongori*) peuvent présenter une couleur magenta peu soutenue en raison d'une activité d'estérase faible.
- Dans le contexte de la validation NORDVAL, le protocole d'enrichissement double a été validé avec un temps d'enrichissement RVS de 24 ± 2 hr seulement.
- Après l'étape d'incubation, la boîte peut être stockée à 2–8 °C pendant 72 hr. Certaines souches non-*Salmonella* (par exemple, *E. coli*) peuvent produire des colonies roses après ce stockage. Ces colonies douteuses donnent un résultat négatif au test de confirmation.

Contrôle qualité

- Chaque produit fabriqué et commercialisé par Bio-Rad est soumis à une procédure d'assurance qualité à toutes les étapes, de la réception des matières premières jusqu'à la mise sur le marché du produit fini. Chaque lot de produits finis subit un contrôle qualité conforme à EN ISO 11133 et est mis sur le marché uniquement s'il satisfait aux critères d'acceptabilité. La documentation relative à la production et au contrôle qualité de chaque lot est archivée.
- Pour consulter la fiche de données de sécurité (FDS) et le certificat d'analyse, visiter www.bio-rad.com

Section 7 Protocole

Préparation du milieu de culture déshydraté

1. Toujours agiter le flacon avant utilisation.
2. Dissoudre 43,5 g de poudre dans 1 L d'eau distillée et mélanger jusqu'à obtenir une suspension homogène.
3. Chauffer doucement en mélangeant fréquemment, puis porter à ébullition pendant moins d'1 min.
4. Ne pas prolonger le chauffage. **Ne pas soumettre à l'autoclave.**
5. Faire refroidir le milieu de culture à 50 °C. Répartir dans les boîtes de Pétri et laisser sécher.
6. 500 g de poudre permettent de reconstituer 11,5 L de milieu.

Préparation du supplément sélectif

1. Une capsule (3564710) est prête à l'emploi pour 25 g ou 25 ml d'échantillon dilué dans 225 ml d'EPT. Il est possible d'ajouter à la poche la capsule entière ou son contenu seulement, avant l'étape de mélange. Pour faciliter la manipulation, nous recommandons d'ouvrir la capsule et de verser son contenu.
2. Les capsules concentrées (10x) (3564709) doivent être diluées avant utilisation. Verser le contenu d'une capsule dans 10 ml d'EPT ou d'eau distillée stérile. Homogénéiser pour obtenir une solution concentrée rouge. Utiliser 1 ml de la solution concentrée pour 25 g ou 25 ml d'échantillon dilué dans 225 ml d'EPT. Pour les échantillons de grande taille, ajuster le volume de supplément sélectif au volume du mélange d'enrichissement pour obtenir une concentration finale de 4 :1000 g/ml. La solution diluée peut être stockée pendant 1 semaine à température ambiante ou à 2–8 °C.
3. Le supplément en boîte (3564712) est concentré et doit être dilué avant utilisation. Ouvrir la boîte et remplir avec 100 ml d'EPT ou d'eau distillée stérile. Homogénéiser en agitant vigoureusement afin d'obtenir une solution concentrée rouge. Utiliser 1 ml de la solution concentrée pour 25 g ou 25 ml d'échantillon dilué dans 225 ml d'EPT. Pour les échantillons de grande taille, ajuster le volume de supplément sélectif au volume du mélange d'enrichissement pour obtenir une concentration finale de 4 :1000 g/ml. La solution diluée peut être stockée pendant 1 semaine à température ambiante ou à 2–8 °C.

- Ouvrir le flacon de PIF Supplement (12013322) et ajouter 40 ml de solution 1:1 éthanol/eau distillée stérile. Homogénéiser en agitant vigoureusement pour obtenir une solution concentrée bleue. Se référer à la notice d'utilisation (document #10000126749) pour les conditions de stockage de la solution diluée.

Enrichissement de l'échantillon

Domaine (matrices)	Enrichissement	Certification
Poitrine de poulet crue, œufs, melon et beurre de cacahouète	n g/ml d'échantillon dans $9 \times n$ ml d'EPT Ajouter RAPID' <i>Salmonella</i> Supplement $41,5 \pm 1$ °C pendant 16–22 hr	AOAC
Large gamme d'aliments, aliments pour animaux et échantillons environnementaux, 25 g	n g/ml d'échantillon dans $9 \times n$ ml d'EPT Ajouter RAPID' <i>Salmonella</i> Supplement $41,5 \pm 1$ °C pendant 16–22 hr	NF VALIDATION
Substitut de lait maternel et ingrédients déshydratés issus du lait, protocole pour échantillon de grande taille, jusqu'à 375 g	n g/ml d'échantillon dans $9 \times n$ ml d'EPT préchauffée (37 °C) (pas de supplément) 37 ± 1 °C pendant 20–24 hr	NF VALIDATION
Large gamme d'aliments, notamment substitut de lait maternel avec ou sans probiotiques, aliments pour animaux et échantillons environnementaux	n g/ml d'échantillon dans $9 \times n$ ml d'EPT Ajouter RAPID' <i>Salmonella</i> Supplement $41,5 \pm 1$ °C pendant 16–22 hr	NORDVAL
Produits laitiers, sauf lait cru	n g/ml d'échantillon dans $9 \times n$ ml d'EPT (pas de supplément) 37 ± 1 °C pendant 16–22 hr Transférer 0,5 ml dans 10 ml de RVS préchauffé $41,5 \pm 1$ °C pendant 6–26 hr	NF VALIDATION
Lait infantiles et céréales infantiles avec et sans probiotiques et ingrédients, jusqu'à 375 g	n g/ml d'échantillon dans $3 \times n$ ml d'EPT préchauffée (37 °C) Ajouter PIF Supplement 37 ± 1 °C pendant 18–24 hr	NF VALIDATION
Aliments pour animaux domestiques et alimentation animale, jusqu'à 375 g	n g/ml d'échantillon dans $5 \times n$ ml d'EPT préchauffée (37 °C) Ajouter RAPID' <i>Salmonella</i> Supplement 37 ± 1 °C pendant 18–24 hr	NF VALIDATION

- Diluer n g ou n ml de l'échantillon dans le volume d'EPT nécessaire selon le taux de dilution utilisé. Voir le tableau ci-dessous.
- Homogénéiser avec un agitateur/homogénéisateur.
- Ajouter le volume requis de supplément sélectif. Le volume de supplément sélectif doit être adapté au volume du mélange d'enrichissement (échantillon + bouillon).
- Incuber conformément au tableau d'enrichissement de l'échantillon.
- Si la méthode de la norme EN ISO 6579-1 est mise en œuvre, suivre le protocole d'enrichissement de l'échantillon tel qu'énoncé.
- Après l'étape d'enrichissement, le bouillon peut être stocké à 2–8 °C pendant 72 hr.

Inoculation et lecture

1. À l'aide d'une anse stérile, strier 10 µl de bouillon d'enrichissement sur RAPID'*Salmonella* Agar.
2. Incuber les boîtes à 37 ± 1 °C pendant 24 ± 2 hr.
3. *Salmonella* spp. forme des colonies magenta sur RAPID'*Salmonella* Agar.
4. Après l'étape d'incubation, les boîtes peuvent être stockées à 2–8 °C pendant 72 hr.

Section 8

Confirmation des résultats positifs

1. Dans le contexte de la validation AOAC, confirmer les colonies isolées suspectes conformément à la procédure de test de confirmation classique décrite dans les méthodes de référence normalisées.
2. Dans le contexte des marques NF VALIDATION et NORDVAL, tous les échantillons identifiés comme positifs doivent être confirmés de l'une des façons suivantes :
 - a. À l'aide des tests classiques décrits dans les méthodes normalisées CEN ou ISO (y compris l'étape de purification).
 - b. À l'aide de sondes nucléiques comme décrit dans la norme ISO 7218 (par exemple, iQ-Check *Salmonella* II Real-Time PCR Kit, n° de référence 3578123) avec des colonies isolées (avec ou sans étape de purification).
 - c. Par évaluation de l'activité d'oxydase, avec un test d'oxydase suivi d'un test antisérum omnivalent Omni-O test (A-60) (*Salmonella* Omni-O Antiserum (A-60), n° de référence 3560781B) avec une à trois colonies suspectes isolées. Si la réaction au test Omni-O est positive, poursuivre avec un test biochimique ONPG (n° de référence 3553822). Les *Salmonella* sont négatives au test d'oxydase, positives au test Omni-O (A-60) et négatives au test ONPG, à l'exception des *Salmonella* lactose positives, qui sont ONPG positives.
 - d. Exécution d'un test d'agglutination au latex sur une colonie isolée à l'aide de *Salmonella* Latex Test (n° de référence 3556710) ou de *Salmonella* Confirm Latex Test (n° de référence 3556711). Le test Oxoid's *Salmonella* Latex a également été validé. Les *Salmonella* appartenant aux sérogroupes B à E et G sont positives avec le test au latex de *Salmonella* Latex Test (n° de référence 3556710).
 - e. Utilisation du test MALDI Biotyper (le système Biotyper de Bruker inclut un spectromètre de masse à temps de vol MALDI et MBT Compass version logicielle 4) directement à partir d'une colonie isolée ou après une étape de purification. Les informations d'identification sur le genre et/ou les espèces fournies par la solution complète MALDI Biotyper ne sont pas incluses dans le cadre de la marque NF VALIDATION.
 - f. À l'aide de toute autre méthode certifiée NF VALIDATION reposant sur un principe différent de celui de RAPID'*Salmonella* Agar. Le protocole validé de cette seconde méthode doit être suivi dans sa totalité. Toutes les étapes précédant l'étape de détection utilisée comme point de départ pour la confirmation doivent être communes aux deux méthodes.
3. En cas de résultats discordants (présumé positif avec RAPID'*Salmonella*, négatif avec la méthode de confirmation, particulièrement avec le test au latex), le laboratoire doit suivre les étapes nécessaires pour garantir la validité du résultat obtenu.

Section 9

Confirmation d'autres méthodes

Confirmation des résultats positifs avec le kit iQ-Check *Salmonella* II (NF VALIDATION – Protocole certifié)

1. Ajouter 500 µl d'EPT enrichie à 10 ml de bouillon RVS préchauffé.
2. Incuber à $41,5 \pm 1$ °C pendant 6–26 hr.
3. À l'aide d'une anse stérile, strier 10 µl de bouillon d'enrichissement sur RAPID'*Salmonella* Agar.
4. Incuber les boîtes à 37 ± 1 °C pendant 24 ± 2 hr.
5. *Salmonella* spp. forme des colonies magenta sur RAPID'*Salmonella* Agar. Procéder à une confirmation sur une colonie magenta à l'aide d'un test d'agglutination au latex ou des tests classiques décrits dans la section 8.2.

Section 10

Performance du test et validations

Certification	Domaine	Protocole de validation	Protocole de référence	Référence de certificat
AOAC-RI	Poitrine de poulet crue, œufs, melon et beurre de cacahouète	Performance Tested Methods	FDA BAM Chap. 5 USDA MLG 4.10	 License n° 050701
NORDVAL	Large gamme d'aliments, notamment substitut de lait maternel avec ou sans probiotiques, aliments pour animaux et échantillons environnementaux	ISO 16140-2:2016	EN ISO 6579:2002	 NORDVAL n° 032
NF VALIDATION	Large gamme d'aliments, aliments pour animaux et échantillons environnementaux (25 g) Lait infantiles et ingrédients déshydratés issus du lait, protocole pour échantillon de grande taille (jusqu'à 375 g), dilution au 1:10 Lait infantiles et céréales infantiles avec et sans probiotiques et ingrédients, protocole	EN ISO 16140-2	EN ISO 6579-1	 BRD : 07/11–12/05 MÉTHODES ANALYTIQUES ALTERNATIVES POUR LE SECTEUR AGRO-ALIMENTAIRE Certifié par AFNOR Certification http://nf-validation.afnor.org/en

Section 11

	pour échantillon de grande taille (jusqu'à 375 g), dilution au 1:4 Aliments pour animaux domestiques et alimentation animale, protocole pour échantillon de grande taille (jusqu'à 375 g), dilution au 1:6			
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Section 11

Références

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United States Department of Agriculture. *Microbiology Laboratory Guidebook*. Chapter 4.10 Isolation and Identification of *Salmonella* from Meat, Poultry Pasteurized Egg, and Siluriformes (Fish) Products and Carcass and Environmental Sponges. <https://www.fsis.usda.gov/wps/wcm/connect/700c05fe-06a2-492a-a6e1-3357f7701f52/mlg-4.pdf?MOD=AJPERES>

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Section 12

Historique des révisions

Date de publication	Numéro de document	Modification
Octobre 2020	10000126748 Ver A	- Modification importante - Nouvelle conception de document - Modification du numéro de document (version précédente RAPID_Salmonella_V11_25 juin 2018)
Juin 2021	10000126748 Ver B	Supplément nouveau format
Janvier 2023	10000126748 Ver C	- Retrait des formats comprimés du supplément - Mise à jour de la préparation du supplément sélectif - Extension NF Validation pour aliments pour animaux domestiques et alimentation animale, jusqu'à 375 g

Visitez www.bio-rad.com/rapidmedia pour plus d'informations sur la gamme complète de milieux chromogènes RAPID.

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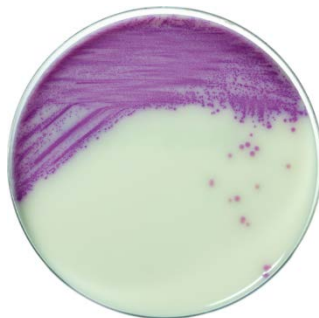
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RAPID'*Salmonella* Agar

Anwenderhandbuch

Chromogenes Medium für den Nachweis von *Salmonella* spp. bei der Untersuchung von Lebensmitteln für den menschlichen Verzehr, Tierfuttermitteln und Umgebungsproben

Katalog-Nr. 3563961, gebrauchsfertige Agarplatten, 20 Agarplatten x 90 mm
Katalog-Nr. 3563963, gebrauchsfertige Agarplatten, 120 Agarplatten x 90 mm
Katalog-Nr. 3564705, dehydriert, 500 g



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Inhaltsverzeichnis

Abschnitt 1	Einleitung	1
Abschnitt 2	Prinzip von RAPID' <i>Salmonella</i>	1
Abschnitt 3	Theoretische Zusammensetzung	1
Abschnitt 4	Haltbarkeit und Lagerung	1
Abschnitt 5	Zusätzlich benötigtes Material	2
	Geräte	2
	Zubehör	2
Abschnitt 6	Vorsichtsmaßnahmen, Anwendungsbeschränkungen und Qualitätskontrolle.....	3
Abschnitt 7	Protokoll	5
	Vorbereitung des dehydrierten Mediums.....	5
	Vorbereitung des selektiven Supplements.....	5
	Probenanreicherung.....	6
	Beimpfung und Ablesen der Platten.....	6
Abschnitt 8	Bestätigung positiver Ergebnisse	8
Abschnitt 9	Bestätigung anderer Methoden	9
Abschnitt 10	Testleistung und Testvalidierungen	10
Abschnitt 11	Literatur	11
Abschnitt 12	Revisionshistorie	12

Abschnitt 1

Einleitung

Salmonella ist ein gramnegatives, nicht sporenbildendes, stäbchenförmiges Bakterium, das bis auf wenige Ausnahmen überwiegend motil ist. *Salmonella* ist eine der beiden häufigsten berichteten Ursachen für bakterielle, lebensmittelbedingte Erkrankungen. Typische Symptome einer Salmonellose sind Durchfall, Fieber und Abdominalkrämpfe, die 12 bis 72 Stunden nach der Infektion auftreten. Studien haben ergeben, dass für jeden im Labor bestätigten Fall von *Salmonella* 38 Fälle im Umfeld des Falles vorliegen, die nicht gemeldet werden. Daher haben Tests auf das Vorhandensein von *Salmonella* in Lebensmitteln und im Umfeld der Verarbeitung einen hohen Stellenwert. Chromogene Substrate in Kulturmedien erleichtern die Verwendung der Medien mit Ergebnissen, die auf enzymatischen Reaktionen mit Farbumschlag beruhen.

Abschnitt 2

Prinzip von RAPID'*Salmonella*

RAPID'*Salmonella* Agar ermöglicht die vorläufige Identifizierung von motilen und nicht-motilen *Salmonella* spp. durch den Nachweis der C8-Esterase-Aktivität. Das gleichzeitige Screening auf die β -Glucosidase-Aktivität ermöglicht die Differenzierung von *Salmonella*-Kolonien von jenen anderer *Enterobacteriaceae*. Nach der Inkubation bildet *Salmonella* gut erkennbare, typischerweise magentafarbene Kolonien, wohingegen andere Bakterien blaue oder farblose Kolonien bilden. RAPID'*Salmonella* Agar ermöglicht den Nachweis von motilen und nicht-motilen *Salmonella* sowie der Laktose-positiven *Salmonella* Typhi und *Salmonella* Paratyphi.

Abschnitt 3

Theoretische Zusammensetzung

Nährstoffmischung	14,5 g
Selektives Agens	14 g
Chromogenes Substrat	2,3 g
Agar	12,7 g
Destilliertes Wasser	qsp 1.000 ml

Finaler pH-Wert bei 25°C = $7,2 \pm 0,2$

Abschnitt 4

Haltbarkeit und Lagerung

- Dehydrierter Agar: Trocken und lichtgeschützt in der sorgfältig verschlossenen Packung bei 2–8°C.
- Gebrauchsfertige Agarplatten: Lichtgeschützt bei 2–8°C.
- Aus dehydriertem Agar hergestellte Agarplatten: 15 Tage bei trockener und lichtgeschützter Lagerung in der sorgfältig verschlossenen Packung bei 2–8°C.

Abschnitt 5

Zusätzlich benötigtes Material

Geräte

- Alle üblichen Laborgeräte
- Heizplatte
- Waage, bis auf 0,1 g genau
- Rührer / Homogenisator
- Thermostatisch kontrollierter Inkubator oder Inkubationskammer, bis auf $\pm 1^{\circ}\text{C}$ genau
- Wasserbad

Zubehör

- Anreicherungsmedium: Gepuffertes Peptonwasser (GPW; Buffered Peptone Water, BPW) (Katalog-Nr. 3554179, 6 Flaschen x 225 ml; 3564684, dehydriert 500 g; 3555790, 2 Beutel x 5 L; 3555795, 4 Beutel x 3 L)
- Impfösen
- PIF Supplement (Katalog-Nr. 12013322, 2 g)
- RAPID'*Salmonella* selektives Supplement für das Kurzprotokoll (Katalog-Nr. 3564709, 100 Kapseln mit 10-fach konzentrierter Lösung; Katalog-Nr. 3564710, 100 Kapseln mit 1-fach konzentrierter Lösung; Katalog-Nr. 3564712, 1 Box x 100 Tests)
- Rappaport Vassiliadis Soja (RVS) für das Protokoll mit doppelter Anreicherung (Katalog-Nr. 3555773, 25 Röhrchen x 10 ml; 3564324, dehydriert, 500 g)
- Sterile Petrischalen
- Sterile Pipetten
- Sterile Probenbeutel

Abschnitt 6

Vorsichtsmaßnahmen, Anwendungsbeschränkungen und Qualitätskontrolle

Vorsichtsmaßnahmen

- Es sind die Richtlinien der guten Laborpraxis zu beachten (EN ISO 7218). Bei der Arbeit mit potenziell infektiösen, lebenden Bakterien wie *Salmonella* sollten angemessene Schutzvorkehrungen getroffen werden (zum Beispiel Handschuhe und Laborkittel tragen).
- Medien, die mit Lebensmittelproben in Kontakt gekommen sind, sind als kontaminiert zu betrachten und gemäß den vor Ort geltenden Vorschriften und Bestimmungen zu entsorgen.
- Für die Zugabe einer ganzen Kapsel mit selektivem Supplement zu einem Beutel mit BPW muss eine sterile Pinzette verwendet werden. (Wir empfehlen zu überprüfen, ob sich die Kapsel tatsächlich öffnet, sodass sich der Inhalt auflösen kann.) Wenn der Kapselinhalt mit bloßen Händen angefasst wird, ist eine Zugabe zu der Anreicherungsbouillon aufgrund des Kontaminationsrisikos nicht mehr möglich.
- Die RAPID'*Salmonella* Kapseln und das RAPID'*Salmonella* Tablette enthalten selektive Agenzien und einen Hilfsstoff. Die selektiven Agenzien lösen sich sehr gut auf. Der Hilfsstoff bleibt jedoch in Suspension und kann einen Niederschlag bilden, wenn der Inhalt der Kapsel in einer kleinen Menge GPW oder sterilem destilliertem Wasser verdünnt wird. Daher muss die konzentrierte Lösung vor dem Gebrauch stets gut geschüttelt werden.

Anwendungsbeschränkungen

- Der ONPG-Bestätigungstest schließt die Bestätigung von Laktose-positiven *Salmonella* aus.
- Mit dem *Salmonella* Confirm Latex Kit können die am häufigsten vorkommenden *Salmonella*-Stämme nachgewiesen werden. Bei der Verlängerung der NF VALIDATION des RAPID'*Salmonella* wurden allerdings 41 der 150 getesteten Stämme mit dem Kit nicht erkannt.
- Manche *Salmonella*-Stämme (von denen einige dem *Salmonella*-Serovar Dublin zuzuordnen sind, sowie *S. bongori*-Spezies) können aufgrund der geringen Esteraseaktivität eine schwache Magentaefärbung aufweisen.
- Im Rahmen der NORDVAL-Validierung wurde das Protokoll mit doppelter Anreicherung mit einer RVS-Anreicherungszeit von nur 24 ± 2 hr validiert.
- Nach der Inkubation können die Platten für 72 hr bei 2–8°C gelagert werden. Einige andere Stämme als *Salmonella* (zum Beispiel *E. coli*) können während der Lagerung unter diesen Bedingungen rosafarbene Kolonien bilden. Diese fraglichen Kolonien sind im Bestätigungstest als negativ zu werten.

Qualitätskontrolle

- Jedes von der Firma Bio-Rad hergestellte und verkaufte Produkt unterliegt vom Rohstoffeingang bis zur Vermarktung der Fertigprodukte einer umfassenden Qualitätssicherung. Jede Charge des fertigen Produkts wird einer Qualitätskontrolle gemäß EN ISO 11133 unterzogen und gelangt nur dann in den Vertrieb, wenn sie die Akzeptanzkriterien erfüllt. Die Dokumente im Zusammenhang mit der Herstellung und der Überprüfung jeder Charge werden archiviert.
- Das Sicherheitsdatenblatt und das Analysezertifikat für das Produkt sind im Internet auf www.bio-rad.com erhältlich.

Abschnitt 7 Protokoll

Vorbereitung des dehydrierten Mediums

1. Den Behälter vor jedem Gebrauch schütteln.
2. 43,5 g Pulver werden in 1 L destilliertem Wasser gelöst und zu einer homogenen Suspension gemischt.
3. Unter ständigem Rühren vorsichtig erhitzen und für weniger als 1 min zum Kochen bringen.
4. Nicht länger erhitzen. **Nicht autoklavieren.**
5. Das Medium auf 50°C abkühlen lassen. In Petrischalen geben und auf dem Tisch trocknen lassen.
6. 500 g Pulver ergeben 11,5 L Medium.

Vorbereitung des selektiven Supplements

1. Eine gebrauchsfertige Kapsel (Katalog-Nr. 3564710) kann direkt für 25 g bzw. 25 ml Probe, verdünnt in 225 ml GPW, verwendet werden. Es kann entweder die gesamte Kapsel oder ihr Inhalt vor dem Mischen in den Probenbeutel gegeben werden. Der Einfachheit halber empfehlen wir, die Kapsel zu öffnen und den Inhalt in den Beutel zu geben.
2. Kapseln mit 10-Fach-Konzentrat (Katalog-Nr. 3564709) müssen vor dem Gebrauch verdünnt werden. Dazu den Inhalt einer Kapsel in 10 ml GPW oder steriles destilliertes Wasser geben. Homogenisieren, um ein rotes, flüssiges Konzentrat zu erhalten. 1 ml konzentrierte Lösung für 25 g bzw. 25 ml Probe, verdünnt in 225 ml GPW, verwenden. Für große Probenvolumina, bitte das Volumen des selektiven Supplements an den Anreicherungsmix anpassen, um eine Konzentration von 4:1000 g/ml zu erreichen. Die verdünnte Lösung kann 1 Woche bei Umgebungstemperatur oder bei 2–8°C gelagert werden.
3. Das Supplement in der Box (Katalog-Nr. 3564712) ist konzentriert und muss vor Gebrauch verdünnt werden. Die Box öffnen und 100 ml GPW oder steriles destilliertes Wasser zugeben. Durch kräftiges Rühren homogenisieren, um ein rotes flüssiges Konzentrat zu erhalten. 1 ml konzentrierte Lösung für 25 g bzw. 25 ml Probe, verdünnt in 225 ml GPW, verwenden. Für große Probenvolumina, bitte das Volumen des selektiven Supplements an den Anreicherungsmix anpassen, um eine Konzentration von 4:1000 g/ml zu erreichen. Die verdünnte Lösung kann 1 Woche bei Umgebungstemperatur oder bei 2–8°C gelagert werden.
4. Die Flasche mit dem PIF Supplement (Katalog Nr. 12013322) öffnen und 40 ml einer 1:1 Lösung aus Ethanol und sterilem, destilliertem Wasser hinzufügen. Durch kräftiges Schütteln mischen um eine blaue, konzentrierte Lösung zu erhalten. Informationen zur Lagerung der verdünnten Lösung sind dem PIF Supplement Anwenderhandbuch (dokument #10000126749) zu entnehmen.

Probenanreicherung

Umfang (Matrizes)	Anreicherung	Zertifizierungsstelle
Rohe Hühnerbrust, Eier, Melone und Erdnussbutter	n g/ml Probe in 9 x n ml GPW Zugabe RAPID' <i>Salmonella</i> Supplement $41,5 \pm 1^\circ\text{C}$ für 16–22 hr	AOAC
Ein breites Spektrum an Nahrungs- und Tierfuttermitteln und Umgebungsproben, 25 g	n g/ml Probe in 9 x n ml GPW Zugabe RAPID' <i>Salmonella</i> Supplement $41,5 \pm 1^\circ\text{C}$ für 16–22 hr	NF VALIDATION
Säuglingsnahrung und zugehörige Milchprodukte in dehydrierter Form zum Zugeben, Protokoll für große Probenmengen, bis zu 375 g	n g/ml Probe in 9 x n ml vorgewärmten (37°C) GPW (Ohne Supplement) $37 \pm 1^\circ\text{C}$ für 20–24 hr	NF VALIDATION
Ein breites Spektrum an Nahrungsmitteln, einschließlich Säuglingsnahrung mit und ohne Probiotika, Tierfutter und Umgebungsproben	n g/ml Probe in 9 x n ml GPW Zugabe RAPID' <i>Salmonella</i> Supplement $41,5 \pm 1^\circ\text{C}$ für 16–22 hr	NORDVAL
Milchprodukte, ausgenommen Rohmilch	n g/ml Probe in 9 x n ml GPW (Ohne Supplement) $37 \pm 1^\circ\text{C}$ für 16–22 hr 0,5 ml in 10 ml vorgewärmtes RVS geben $41,5 \pm 1^\circ\text{C}$ für 6–26 hr	NF VALIDATION
Säuglingsnahrung und Kinderzerealien mit und ohne Probiotika und Zusatzstoffen, bis zu 375 g	n g/ml Probe in 3 x n ml vorgewärmten (37°C) GPW Zugabe PIF Supplement $37 \pm 1^\circ\text{C}$ für 18–24 hr	NF VALIDATION
Heimtierfutter und Tiernahrung, bis zu 375 g	n g/ml Probe in 5 x n ml vorgewärmten (37°C) GPW Zugabe RAPID' <i>Salmonella</i> Supplement $37 \pm 1^\circ\text{C}$ für 18–24 hr	NF VALIDATION

1. n g oder n ml der Probe im erforderlichen Volumen an GPW gemäß dem entsprechenden Verdünnungsverhältnis verdünnen. Siehe Tabelle.
2. Mit einem Rührer/Homogenisator homogenisieren.
3. Das erforderliche Volumen an selektivem Supplement zugeben. Das Volumen des selektive Supplements muss an das Volumen des Anreicherungsmix angepasst werden (Probe + Peptonwasser).
4. Nach den Angaben in der Tabelle zur Probenanreicherung inkubieren.
5. Bei der Durchführung der Methode nach EN ISO 6579-1 ist das in der Norm beschriebene Protokoll zur Probenanreicherung zu befolgen.
6. Nach der Anreicherung kann die Bouillon 72 hr bei $2\text{--}8^\circ\text{C}$ gelagert werden.

Beimpfung und Ablesen der Platten

1. Mit einer sterilen Impföse 10 μl Anreicherungsbouillon auf RAPID'*Salmonella* Agar ausstreichen.

Abschnitt 7 Protokoll

2. Die Platten bei $37 \pm 1^\circ\text{C}$ für 24 ± 2 hr inkubieren.
3. *Salmonella* spp. bildet auf RAPID'*Salmonella* Agar magentafarbene Kolonien.
4. Nach der Inkubation können die Platten für 72 hr bei $2-8^\circ\text{C}$ gelagert werden.

Abschnitt 8

Bestätigung positiver Ergebnisse

1. Im Rahmen einer AOAC-Validierung vermutete Einzelkolonien nach dem klassischen Bestätigungstestverfahren, das in den Standardreferenzmethoden beschrieben ist, bestätigen.
2. Im Rahmen des Verfahrens zum Erhalt des Zeichens für die NF VALIDATION und NORDVAL-Validierung müssen alle als positiv identifizierten Proben mit einer der folgenden Methoden bestätigt werden:
 - a. Mit den in CEN- oder ISO-Normen beschriebenen herkömmlichen Tests (einschließlich des Reinigungsschritts).
 - b. Durch Verwendung von Nukleinsonden wie in der Norm ISO 7218 beschrieben (z. B. iQ-Check *Salmonella* II Real-Time PCR Kit, Katalog-Nr. 3578123) unter Verwendung von Einzelkolonien (mit oder ohne Reinigungsschritt).
 - c. Durch Bestimmung der Oxidaseaktivität mit einem Oxidasetest gefolgt von einem Omnivalent Omni-O-Test (A-60) (*Salmonella* Omni-O Antiserum (A-60) Katalog-Nr. 3560781B) unter Verwendung von 1 bis 3 isolierten mutmaßlichen Kolonien. Bei einem positiven Ergebnis im Omni-O-Test mit einem biochemischen ONPG-Test (Katalog-Nr. 3553822) fortfahren. *Salmonella* sind negativ im Oxidasetest, positiv im Omni-O-Test (A-60) und negativ im ONPG-Test, mit Ausnahme von Laktose-positiven *Salmonella*, die ONPG-positiv sind.
 - d. Mit einem Latexagglutinationstests unter Verwendung des *Salmonella* Latex Tests (Katalog-Nr. 3556710) mit einer isolierten Kolonie oder mit dem *Salmonella* Confirm Latex Test (Katalog-Nr. 3556711) mit einer isolierten Kolonie. Der *Salmonella* Latex Test von Oxoid wurde ebenfalls validiert. *Salmonella* der Serogruppen B bis E und G sind im Latex-Test positiv des *Salmonella* Latex Tests (Katalog-Nr. 3556710).
 - e. Durch Anwendung des MALDI Biotyper Tests (das Biotyper-System von Bruker umfasst ein MALDI-Flugzeit-Massenspektrometer und die MBT Compass-Software, Version 4) direkt mit einer isolierten Kolonie oder nach einem Reinigungsschritt. Die mit der MALDI Biotyper Complete Solution erhaltenen Informationen zur Identifizierung der Gattung und/oder Art werden im Zusammenhang mit der Erteilung der NF VALIDATION-Zertifizierung nicht berücksichtigt.
 - f. Durch Verwendung einer anderen NF VALIDATION-zertifizierten Methode, die auf einem anderen Testprinzip als dem von RAPID'*Salmonella* Agar beruht. Das validierte Protokoll dieser zweiten Methode muss ohne Abwandlungen durchgeführt werden. Alle Schritte vor dem Nachweissschritt als Ausgangspunkt für die Bestätigung müssen in beiden Methoden identisch sein.
3. Bei nicht übereinstimmenden Ergebnissen (vermutlich positiv auf RAPID'*Salmonella*, negativ mit der Bestätigungsmethode und insbesondere beim Latex-Test) muss das Labor die erforderlichen Schritte ausführen, um die Validität der erhaltenen Ergebnisse sicherzustellen.

Abschnitt 9

Bestätigung anderer Methoden

Bestätigung positiver Ergebnisse mit dem iQ-Check *Salmonella* II Kit (von NF VALIDATION zertifiziertes Protokoll)

1. 500 µl angereichertes GPW zu 10 ml vorgewärmter RVS-Bouillon geben.
2. Bei $41,5 \pm 1^\circ\text{C}$ für 6 bis 26 hr inkubieren.
3. Mit einer sterilen Impföse 10 µl Anreicherungsbouillon auf RAPID'*Salmonella* Agar ausstreichen.
4. Die Platten bei $37 \pm 1^\circ\text{C}$ für 24 ± 2 hr inkubieren.
5. *Salmonella* spp. bildet auf RAPID'*Salmonella* Agar magentafarbene Kolonien. Eine Bestätigung einer magentafarbenen Kolonie mit einem Latexagglutinationstest oder mit den in Abschnitt 8.2 beschriebenen herkömmlichen Tests durchführen.

Abschnitt 10

Testleistung und Testvalidierungen

Zertifizierungsstelle	Umfang	Validierungsprotokoll	Referenzprotokoll	Zertifikat-Referenz
AOAC-RI	Rohe Hühnerbrust, Eier, Melone und Erdnussbutter	Leistungsgeprüfte Methoden	FDA BAM Chap. 5 USDA MLG 4.10	 Lizenznr. 050701
NORDVAL	Ein breites Spektrum an Nahrungsmitteln, einschließlich Säuglingsnahrung mit und ohne Probiotika, Tierfutter und Umgebungsproben	ISO 16140-2:2016	EN ISO 6579:2002	 NORDVAL # 032
NF VALIDATION	Breites Spektrum an Nahrungs- und Tierfuttermitteln und Umgebungsproben (25 g) Säuglingsnahrung und zugehörige Milchprodukte in dehydrierter Form zum Zugeben, Protokoll für große Probenmengen (bis zu 375 g), 1:10 Verdünnung Säuglingsnahrung und Kinderzerealien mit und ohne Probiotika und Zusatzstoffen, Protokoll für große Probenmengen (bis zu 375 g), 1:4 Verdünnung Heimtierfutter und Tiernahrung, Protokoll für große Probenmengen (bis zu 375 g), 1:6 Verdünnung	EN ISO 16140-2	EN ISO 6579-1	 BRD: 07/11-12/05 ALTERNATIVE ANALYTICAL METHODS FOR AGRIBUSINESS Zertifiziert durch die AFNOR-Zertifizierungsstelle http://nf-validation.afnor.org/en

Abschnitt 11 Literatur

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Abschnitt 12

Revisionshistorie

Freigabedatum	Dokumentnummer	Änderung
Oktober 2020	10000126748 Ver A	- Bedeutende Änderung - Neues Dokumentdesign - Änderung der Dokumentnummer - vorhergehende Version RAPID_Salmonella_V11_25. Juni 2018
Juni 2021	10000126748 Ver B	Supplement in neuem Format
Januar 2023	10000126748 Ver C	- Aktualisierung bzgl. Vorbereitung des selektiven Supplements - Erweiterung der NF Validierung für - Heimtierfutter und Tiernahrung bis zu 375 g

Weitere Informationen über unser vollständiges Angebot an chromogenen RAPID Medien finden Sie bei uns im Internet auf www.bio-rad.com/rapidmedia.

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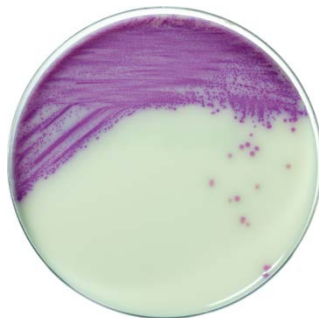
10000126748 Ver C US/EG

RAPID'*Salmonella* Agar

Manuale utente

Terreni cromogenici per la ricerca di *Salmonella* spp. nell'analisi di prodotti alimentari destinati al consumo umano, mangimi per animali e campioni ambientali.

Catalogo #3563961, piastre preparate, 90 mm x 20 piastre
Catalogo #3563963, piastre preparate, 90 mm x 120 piastre
Catalogo #3564705, disidratato, 500 g



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Indice

Sezione 1	Introduzione	1
Sezione 2	RAPID'Salmonella - Principio	1
Sezione 3	Formula teorica	1
Sezione 4	Durata e conservazione	1
Sezione 5	Materiali necessari ma non forniti	2
	Apparecchiatura	2
	Materiali	2
Sezione 6	Precauzioni, limitazioni d'uso e controllo qualità	3
Sezione 7	Protocollo	4
	Preparazione del terreno disidratato	4
	Preparazione del supplemento selettivo.....	4
	Arricchimento del campione	5
	Inoculazione e lettura delle piastre	5
Sezione 8	Conferma dei risultati positivi	7
Sezione 9	Conferma di altri metodi	8
Sezione 10	Performance del test e validazioni	9
Sezione 11	Riferimenti	10
Sezione 12	Cronologia delle revisioni	11

Sezione 1

Introduzione

La *Salmonella* è un batterio Gram negativo, non sporigeno, a forma di bastoncello, prevalentemente motile, con solo alcune eccezioni. La *Salmonella* è stata una delle due cause più frequentemente segnalate di malattia batterica di origine alimentare. I sintomi comuni di salmonellosi includono diarrea, febbre e crampi addominali con insorgenza 12-72 hr dopo l'infezione. Gli studi hanno dimostrato che per ogni caso confermato in laboratorio di *Salmonella*, si verificano 38 casi non segnalati nella comunità. L'esame per rilevare la presenza di *Salmonella* nei prodotti alimentari e nell'ambiente di lavorazione degli alimenti costituisce una pratica importante. L'uso di substrati cromogenici nei terreni di coltura ne aumenta la facilità d'uso grazie a risultati basati su reazioni enzimatiche che mutano di colore.

Sezione 2

RAPID'*Salmonella* - Principio

L'agar RAPID'*Salmonella* consente l'identificazione presuntiva di *Salmonella* spp. motile e non motile rilevando l'attività della C8 esterasi. Lo screening simultaneo dell'attività della β -glucosidasi permette di differenziare le colonie di *Salmonella* da quelle di altre *Enterobacteriaceae*. Dopo l'incubazione, la *Salmonella* si presenta in colonie facilmente identificabili, tipicamente di color magenta, mentre i batteri non *Salmonella* crescono in colonie di colore blu o incolori. L'agar RAPID'*Salmonella* permette di rilevare la *Salmonella* motile e non motile, oltre alla *Salmonella* Typhi e alla *Salmonella* Paratyphi, lattosio-positive.

Sezione 3

Formula teorica

Miscela nutritiva	14,5 g
Agenti selettivi	14 g
Miscela cromogenica	2,3 g
Agar	12,7 g
Acqua distillata	qs 1.000 ml

pH finale a 25°C = 7,2 $\pm$ 0,2

Sezione 4

Durata e conservazione

- Agar disidratato: 2-8°C in una confezione accuratamente sigillata e conservato in un luogo asciutto e buio
- Agar preparato: 2-8°C in un luogo buio
- Piastra preparata da agar disidratato: 15 giorni a 2-8°C in una confezione sigillata attentamente e conservata in un luogo asciutto e buio

Sezione 5

Materiali necessari ma non forniti

Apparecchiatura

- Tutta la normale apparecchiatura di laboratorio
- Piastra riscaldante
- Bilancia, sensibilità di 0,1 g
- Agitatore/omogeneizzatore
- Incubatore o camera di incubazione con controllo termostatico, con precisione di $\pm 1^{\circ}\text{C}$
- Bagnomaria

Materiali

- Terreno di arricchimento: Acqua peptonata tamponata (APT) (catalogo #3554179, 225 ml x 6 bottiglie; 3564684, disidratato 500 g; 3555790, 5 L x 2 sacche; 3555795, 3 L x 4 sacche)
- Anse per inoculazione
- PIF Supplement (catalogo #12013322, 2 g)
- Supplemento selettivo RAPID'*Salmonella* per protocollo breve (catalogo #3564709, 10x concentrato x 100 capsule; n. 3564710, 1x concentrato x 100 capsule; n. 3564712, 1 confezione x 100 test)
- Rappaport Vassiliadis Soy (RVS) per protocollo di arricchimento doppio (catalogo #3555773, 10 ml x 25 provette; 3564324, disidratato 500 g)
- Piastre Petri sterili
- Pipette sterili
- Sacche di pesatura sterili

Sezione 6

Precauzioni, limitazioni d'uso e controllo qualità

Precauzioni

- Rispettare le buone pratiche di laboratorio (EN ISO 7218). Indossare protezioni adeguate, quali guanti e camici da laboratorio, quando si manipolano batteri vivi potenzialmente infettivi come la *Salmonella*
- I terreni entrati in contatto con campioni di prodotti alimentari devono essere considerati contaminati e quindi smaltiti in conformità con normative e direttive locali
- Se l'intera capsula di supplemento selettivo viene aggiunta alla APT, utilizzare pinzette sterili per aggiungere la capsula alla sacca. (Si consiglia di verificare l'apertura della capsula durante la fase di miscelazione.) Se il contenuto della capsula è manipolato con le dita, non può essere aggiunto al brodo di arricchimento per il rischio di contaminazione
- Le capsule RAPID'*Salmonella* e le compresse RAPID'*Salmonella* contengono agenti selettivi e un eccipiente. Gli agenti selettivi si dissolvono facilmente. L'eccipiente, tuttavia, rimane in sospensione e può creare un deposito quando il contenuto della capsula è diluito in una piccola quantità di BPW o di acqua distillata sterile. Quindi, agitare sempre bene prima di utilizzare la soluzione concentrata

Limitazioni d'uso

- Il test di conferma ONPG esclude la conferma di *Salmonella* lattosio-positiva.
- Benché i ceppi più frequenti di *Salmonella* possano essere rilevati dal kit *Salmonella* Confirm Latex, il kit non ha rilevato 41 dei 150 ceppi testati durante l'estensione della validazione NF di RAPID'*Salmonella*
- Qualche ceppo di *Salmonella* (alcuni fanno parte delle specie *Salmonella* serovar Dublin e *S. bongori*) può presentare una leggera colorazione a causa di una bassa attività esterasica
- Nell'ambito della validazione NORDVAL, il protocollo di arricchimento doppio è stato validato con un tempo di arricchimento RVS di solo 24 ± 2 hr
- Al termine della fase di incubazione, la piastra può essere conservata a 2-8°C per 72 hr. Alcuni ceppi non *Salmonella* (ad esempio *E. coli*) possono sviluppare colonie di colore rosa dopo la conservazione. Queste colonie incerte generano un test di conferma negativo

Controllo qualità

- Tutti i prodotti fabbricati e commercializzati dalla società Bio-Rad sono sottoposti a un sistema di assicurazione qualità in ogni fase, dal momento del ricevimento delle materie prime fino alla commercializzazione dei prodotti finiti. Ciascun lotto di prodotto finito è soggetto a un controllo qualità conformemente alla norma EN ISO 11133 e viene messo in commercio solo se risulta conforme ai criteri di accettazione. La documentazione relativa alla produzione e al controllo qualità di ogni lotto è conservata su file
- Per informazioni sulla sicurezza del prodotto (schede dati di sicurezza) e il certificato di analisi, visitare il sito www.bio-rad.com

Sezione 7

Protocollo

Preparazione del terreno disidratato

1. Agitare sempre il flacone prima dell'uso.
2. Dissolvere 43,5 g di polvere in 1 L di acqua distillata e miscelare fino a ottenere una sospensione omogenea.
3. Riscaldare lentamente, agitando spesso, quindi portare a ebollizione per meno di 1 min.
4. Non riscaldare per un tempo superiore. **Non autoclavare.**
5. Raffreddare il terreno a 50°C. Dispensare in piastre di Petri e lasciare asciugare sul piano di lavoro.
6. Un flacone di polvere da 500 g produce 11,5 L di terreno.

Preparazione del supplemento selettivo

1. Una capsula (3564710) è pronta all'uso per 25 g o 25 ml di campione diluito in 225 ml di APT. Prima della fase di miscelazione è possibile aggiungere alla sacca del campione l'intera capsula o il suo contenuto. Per facilitare la manipolazione, si consiglia di aprire la capsula e versare all'interno il contenuto.
2. Le capsule (10x) di concentrato (3564709) devono essere diluite prima dell'uso. Versare il contenuto di una capsula in 10 ml di APT o di acqua distillata sterile. Omogeneizzare fino a ottenere una soluzione concentrata di colore rosso. Utilizzare 1 ml di soluzione concentrata per 25 g o 25 ml di campione diluito in 225 ml di APT. Per campioni di grandi dimensioni, adattare il volume del supplemento selettivo a quello dell'arricchimento per ottenere una concentrazione di 4:1000 g/ml. La soluzione diluita può essere conservata per 1 settimana a temperatura ambiente o a 2-8°C.
3. Il supplemento confezionato (3564712) è concentrato e deve essere diluito prima dell'uso. Aprire la confezione e riempire con 100 ml di APT o di acqua distillata sterile. Omogeneizzare agitando vigorosamente fino a ottenere una soluzione concentrata di colore rosso. Utilizzare 1 ml di soluzione concentrata per 25 g o 25 ml di campione diluito in 225 ml di APTW. Per campioni di grandi dimensioni, adattare il volume del supplemento selettivo a quello dell'arricchimento per ottenere una concentrazione di 4:1000 g/ml. La soluzione diluita può essere conservata per 1 settimana a temperatura ambiente o a 2-8°C.
4. Aprire il contenitore del PIF Supplement (12013322) e riempirlo con 40 ml di una soluzione Etanolo/acqua distillata 1:1. Dissolvere agitando energicamente per ottenere una soluzione concentrata di colore blu. Fare riferimento alla guida utente PIF supplement (documento #10000126749) per informazioni sulla conservazione della soluzione diluita.

Arricchimento del campione

Oggetto (matrici)	Arricchimento	Certificazione
Petto di pollo crudo, uova, melone e burro di arachidi	n g/ml di campione in 9 x n ml APT Aggiungere RAPID' <i>Salmonella</i> Supplement $41,5 \pm 1^\circ\text{C}$ per 16–22 hr	AOAC
Ampia gamma di prodotti alimentari, mangimi per animali e campioni ambientali, 25 g	n g/ml di campione in 9 x n ml APT Aggiungere RAPID' <i>Salmonella</i> Supplement $41,5 \pm 1^\circ\text{C}$ per 16–22 hr	NF VALIDATION
Formula per neonati e relativi ingredienti lattiero-caseari disidratati, protocollo per campioni di grandi dimensioni, fino a 375 g	n g/ml di campione in 9 x n ml APT preriscaldata (37°C) (senza supplemento) $37 \pm 1^\circ\text{C}$ per 20–24 hr	NF VALIDATION
Ampia gamma di prodotti alimentari, inclusa formula per neonati con e senza probiotici, mangimi per animali e campioni ambientali	n g/ml di campione in 9 x n ml APT Aggiungere RAPID' <i>Salmonella</i> Supplement $41,5 \pm 1^\circ\text{C}$ per 16–22 hr	NORDVAL
Prodotti lattiero-caseari, escluso il latte crudo	n g/ml di campione in 9 x n ml APT (senza supplemento) $37 \pm 1^\circ\text{C}$ per 16–22 hr Trasferire 0,5 ml in 10 ml di RVS preriscaldato $41,5 \pm 1^\circ\text{C}$ per 6–26 hr	NF VALIDATION
Preparazione per infanti e cereali per infanti con o senza probiotici ed ingredienti, fino a 375 g	n g/ml di campione in 3 x n ml APT preriscaldata (37°C) Aggiungere PIF Supplement $37 \pm 1^\circ\text{C}$ per 18–24 hr	NF VALIDATION
Alimenti per animali domestici e mangime per animali fino a 375 g	n g/ml di campione in 5 x n ml APT preriscaldata (37°C) Aggiungere RAPID' <i>Salmonella</i> Supplement $37 \pm 1^\circ\text{C}$ per 18–24 hr	NF VALIDATION

1. Diluire n g o n ml di campione nel volume richiesto di APT seguendo il corretto rapporto di diluizione. Consultate la tabella.
2. Omogeneizzare con un agitatore/omogeneizzatore.
3. Aggiungere il volume richiesto di supplemento selettivo. Il volume del supplemento selettivo deve essere adattato a quello dell'arricchimento (campione + BPW).
4. Incubare secondo la tabella di arricchimento del campione.
5. Quando si segue il metodo della EN ISO 6579-1, attenersi al protocollo di arricchimento del campione come descritto dalla norma.
6. Al termine della fase di arricchimento, il brodo può essere conservato a $2-8^\circ\text{C}$ per 72 hr.

Inoculazione e lettura delle piastre

1. Utilizzando un'ansa sterile, strisciare 10 μl di brodo di arricchimento sull'agar RAPID'*Salmonella*.

2. Incubare le piastre a $37 \pm 1^{\circ}\text{C}$ per 24 ± 2 hr.
3. La *Salmonella* spp. forma colonie di color magenta sull'agar RAPID'*Salmonella* .
4. Al termine della fase di incubazione, le piastre possono essere conservate a $2-8^{\circ}\text{C}$ per 72 hr.

Sezione 8

Conferma dei risultati positivi

1. Nell'ambito della validazione AOAC, confermare le colonie isolate sospette secondo la tradizionale procedura del test di conferma descritta nei metodi di riferimento standard.
2. Nell'ambito della VALIDAZIONE NF e NORDVAL, tutti i campioni identificati come positivi devono essere confermati in uno dei seguenti modi:
 - a. Utilizzando i test convenzionali descritti nei metodi standard CEN o ISO (inclusa la fase di purificazione).
 - b. Mediante sonde nucleiche, come descritto nella norma ISO 7218 (ad esempio, il kit iQ-Check *Salmonella* II Real-Time PCR, catalogo #3578123) utilizzando colonie isolate (con o senza fase di purificazione).
 - c. Valutando l'attività dell'ossidasi con un test dell'ossidasi seguito da un test Omni-O omnivalente (A-60) (*Salmonella* Omni-O Antiserum (A-60) catalogo #3560781B) usando da una a tre colonie sospette isolate. Se la reazione al test Omni-O è positiva, procedere con un test biochimico ONPG (catalogo #3553822). La *Salmonella* è negativa al test dell'ossidasi, positiva al test Omni-O test (A-60) e negativa al test ONPG, ad eccezione della *Salmonella* lattosio-positiva, che è positiva all'ONPG.
 - d. Eseguendo un test di agglutinazione al lattice tramite il Latex Test (catalogo #3556710) *Salmonella* su una colonia isolata, oppure il Confirm Latex Test (catalogo #3556711) *Salmonella* su una colonia isolata. Anche il Latex Test *Salmonella* di Oxoid è stato validato. La *Salmonella* appartenente ai sierogruppi da B a E e G è positiva al test al lattice il Latex Test (catalogo #3556710) *Salmonella*.
 - e. Utilizzando il test MALDI Biotyper (il sistema Biotyper di Bruker include uno spettrometro di massa a tempo di volo MALDI e il software MBT Compass, versione 4) direttamente da una colonia isolata o dopo una fase di purificazione. Le informazioni sull'identificazione del genere e/o della specie fornite dalla soluzione completa MALDI Biotyper non sono incluse nell'ambito del marchio NF VALIDATION.
 - f. Usando qualsiasi altro metodo certificato di VALIDAZIONE NF basato su un principio diverso rispetto a quello dell'agar RAPID' *Salmonella*. Il protocollo validato del secondo metodo deve essere rispettato interamente. Tutte le fasi che precedono la fase di ricerca usata come punto di inizio per la conferma devono essere comuni a entrambi i metodi.
3. In caso di risultati discordanti (presunto positivo con RAPID' *Salmonella*, negativo con metodo di conferma, in particolare con il test al lattice), il laboratorio deve seguire le fasi necessarie per garantire la validità dei risultati ottenuti.

Sezione 9

Conferma di altri metodi

Conferma dei risultati positivi con il kit iQ-Check *Salmonella* II (protocollo certificato con VALIDAZIONE NF)

1. Aggiungere 500 µl di APT arricchita a 10 ml di brodo RVS preriscaldato.
2. Incubare a $41,5 \pm 1^{\circ}\text{C}$ per 6–26 hr
3. Utilizzando un'ansa sterile, strisciare 10 µl di brodo di arricchimento sull'agar RAPID'*Salmonella*.
4. Incubare le piastre a $37 \pm 1^{\circ}\text{C}$ per 24 ± 2 hr.
5. La *Salmonella* spp. forma colonie di color magenta sull'agar RAPID'*Salmonella* . Eseguire la conferma di una colonia magenta usando il test di agglutinazione al lattice o i test tradizionali descritti nella Sezione 8.2.

Sezione 10

Performance del test e validazioni

Certificazione	Ambito di applicazione	Protocollo di validazione	Protocollo di riferimento	Riferimento al certificato
AOAC-RI	Petto di pollo crudo, uova, melone e burro di arachidi	Performance Tested Methods	FDA BAM Cap. 5 USDA MLG 4.10	 Licenza n. 050701
NORDVAL	Ampia gamma di prodotti alimentari, inclusa formula per neonati con e senza probiotici, mangimi per animali e campioni ambientali	ISO 16140-2:2016	EN ISO 6579:2002	 NORDVAL n. 032
NF VALIDATION	Ampia gamma di prodotti alimentari, mangimi per animali e campioni ambientali (25 g) Formula per neonati e relativi ingredienti lattiero-caseari disidratati, protocollo per campioni di grandi dimensioni (fino a 375 g), diluizione 1:10 Formula per neonati e cereali per infanti con o senza probiotici ed ingredienti, protocollo per campioni di grandi dimensioni (fino a 375 g), diluizione 1:4 Alimenti per animali domestici e mangime per animali protocollo per campioni di grandi dimensioni (fino a 375 g), diluizione 1:6	EN ISO 16140-2	EN ISO 6579-1	 BRD: 07/11-12/05 METODI ANALITICI ALTERNATIVI PER IL SETTORE AGROALIMENTARE Certificato mediante certificazione AFNOR http://nf-validation.afnor.org/en

Sezione 11

Riferimenti

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Manafi M et al. (1991). Fluorogenic and chromogenic substrates used in bacterial diagnostics. Microbiol Rev, 55, 335–348.

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Sezione 12

Cronologia delle revisioni

Data di pubblicazione	Numero documento	Modifica
Ottobre 2020	10000126748 Ver A	- Modifica importante - Nuovo design del documento - Modifica numero documento – versione precedente RAPID_Salmonella_V11_25 giugno 2018
Giugno 2021	10000126748 Ver B	Nuovo formato del supplemento
Gennaio 2023	10000126748 Ver C	- Ritiro del supplemento nel formato capsule - Aggiornamento nella preparazione del supplemento selettivo - Estensione della validazione NF ad limenti per animali domestici e mangime per animali fino a 375 g

Per ulteriori informazioni sulla nostra gamma completa di terreni cromogenici RAPID', visitare il sito www.bio-rad.com/rapidmedia.

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RAPID'*Salmonella* Agar

Guia do usuário

Meios cromogênicos para a detecção de *Salmonella* spp. na análise de produtos alimentícios para consumo humano, ração animal e em amostras ambientais

Nº do catálogo 3563961, Meios de cultura preparados, 90 mm x 20 placas

Nº do catálogo 3563963, Meios de cultura preparados, 90 mm x 120 placas

Nº do catálogo 3564705, Desidratado, 500 g



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Índice

Seção 1	Introdução	1
Seção 2	Princípio RAPID' <i>Salmonella</i>	1
Seção 3	Fórmula Teórica.....	1
Seção 4	Prazo de validade e armazenamento	1
Seção 5	Materiais necessários, mas não fornecido	2
	Equipamento	2
	Suprimentos.....	2
Seção 6	Precauções, limitações de uso e controle de qualidade	3
Seção 7	Protocolo	4
	Preparação do Meio Desidratado.....	4
	Preparação do suplemento seletivo	4
	Enriquecimento da amostra	5
	Inoculação e Leitura de Meios de Cultura	5
Seção 8	Confirmação de Resultados Positivos.....	6
Seção 9	Confirmação de outros métodos.....	7
Seção 10	Desempenho e validação do teste	8
Seção 11	Referências	9
Seção 12	Histórico de Revisão	10

Seção 1 Introdução

A *Salmonella* é uma bactéria gram-negativa, que não forma esporos, em forma de bastão, que é predominantemente móvel, com poucas exceções. A *Salmonella* tem sido uma das duas causas mais frequentemente relatadas de doenças bacterianas de origem alimentar. Os sintomas comuns da salmonelose incluem diarreia, febre e cólicas abdominais que ocorrem de 12 a 72 hr após a infecção. Estudos demonstraram que para cada caso de *Salmonella* confirmado pelo laboratório, 38 casos ocorrem na comunidade que permanecem não notificados. O teste da presença de *Salmonella* nos alimentos e no ambiente de processamento é uma prática importante. O uso de substratos cromogênicos em meios de cultura aumenta a facilidade de uso dos meios com resultados baseados em reações enzimáticas de mudança de cor.

Seção 2 Princípio RAPID'*Salmonella*

O RAPID'*Salmonella* Agar permite a presumível identificação de *Salmonella* spp. móvel e não móvel através da detecção de atividade C8-esterase. A triagem simultânea da atividade de β -glucosidase permite a diferenciação das colônias de *Salmonella* das de outras *Enterobacteriaceae*. Após a incubação, a *Salmonella* aparece como facilmente identificável, tipicamente magenta, em colônias, enquanto as bactérias não *Salmonella* crescem como colônias azuis ou incolores. RAPID'*Salmonella* Agar permite a detecção de *Salmonella* móvel e não móvel, assim como *Salmonella* Typhi e *Salmonella* Paratyphi positivas à lactose.

Seção 3 Fórmula Teórica

Mistura nutritiva	14,5 g
Agentes seletivos	14 g
Mistura cromogênica	2,3 g
Ágar	12,7 g
Água destilada	qsp 1.000 ml

pH final a 25°C = 7,2 $\pm$ 0,2

Seção 4 Prazo de validade e armazenamento

- Ágar desidratado: 2–8°C em embalagem cuidadosamente vedada e armazenada em um ambiente seco e arejado
- Ágar derramada: 2–8°C em local escuro
- Meio de cultura preparado a partir do ágar desidratado: 15 dias a 2–8°C em embalagem cuidadosamente vedada e armazenada em um ambiente seco e arejado

Seção 5

Materiais necessários, mas não fornecidos

Equipamento

- Todo o equipamento comum de laboratório
- Placa de aquecimento
- Escala, sensibilidade de 0,1 g
- Misturador/homogeneizador
- Incubadora ou sala de incubação controlada termostaticamente, com precisão de $\pm 1^{\circ}\text{C}$
- Lavagem em água

Suprimentos

- Meio de enriquecimento: Água peptonada tamponada (BPW) (nº do catálogo 3554179, 225 ml x 6 frascos; 3564684, desidratado 500 g; 3555790, 5 L x 2 sacos; 3555795, 3 L x 4 sacos)
- Inoculação de loops
- PIF Supplement (nº do catálogo 12013322, 2 g)
- RAPID'*Salmonella* suplemento seletivo para protocolo curto (catálogo #3564709, 10x cápsulas concentradas x 100; 3564710, 1x cápsulas concentradas x 100)
- Rappaport Vassiliadis Soy (RVS) para protocolo de duplo enriquecimento (catálogo #3555773, 10 ml x 25 tubos; 3564324, desidratado 500 g)
- Placas de Petri estéreis
- Pipetas estéreis
- Sacos de pesagem estéreis

Seção 6

Precauções, limitações de uso e controle de qualidade

Precauções

- Respeite as boas práticas de laboratório (EN ISO 7218). É necessário o uso de proteção adequada, como luvas e jalecos, ao trabalhar com bactérias vivas potencialmente infecciosas, como a *Salmonella*
- O meio que entrou em contato com amostras de alimentos deve ser considerado contaminado e descartado de acordo com as regras e regulamentos locais
- Se toda a cápsula do suplemento seletivo for adicionada à BPW, uma pinça esterilizada deve ser usada para adicionar a cápsula ao saco. (Recomendamos que você verifique se a cápsula realmente se abriu durante a fase do estômago.) Se o conteúdo da cápsula for manuseado com os dedos, não poderá ser adicionado ao caldo de enriquecimento devido ao risco de contaminação
- As cápsulas RAPID'*Salmonella* e tabletes RAPID'*Salmonella* contêm agentes seletivos e um excipiente. Os agentes seletivos se dissolvem muito bem. O excipiente, entretanto, permanece em suspensão e pode criar um depósito quando o conteúdo da cápsula é diluído em uma pequena quantidade de BPW ou água destilada estéril. Agite sempre bem, portanto, antes de usar a solução concentrada

Limitações de uso

- O teste de confirmação ONPG exclui a confirmação de *Salmonella* positiva à lactose
- Embora as cepas de *Salmonella* mais prevalentes possam ser detectadas pelo kit de confirmação de látex de *Salmonella*, o kit não detectou 41 das 150 cepas testadas durante a extensão da NF VALIDATION do RAPID'*Salmonella*
- Algumas cepas de *Salmonella* (algumas fazem parte de espécies de *Salmonella* serotipo Dublin e *S. bongori*) podem mostrar uma cor magenta fraca devido à baixa atividade de esterase
- No contexto da validação NORDVAL, o protocolo de duplo enriquecimento foi validado com um tempo de enriquecimento RVS de apenas 24 ± 2 h
- Após o passo de incubação, a placa pode ser armazenada a 2–8°C por 72 h. Algumas cepas não *Salmonella* (por exemplo, *E. coli*) podem desenvolver colônias de cor rosa após este armazenamento. Estas colônias duvidosas dão um teste de confirmação negativo

Controle de qualidade

- Todos os produtos fabricados e comercializados pela Bio-Rad estão sujeitos aos procedimentos de garantia de qualidade em todas as etapas, desde a recepção da matéria-prima até a comercialização do produto final. Cada lote de produto acabado passa por um controle de qualidade de acordo com a EN ISO 11133 e é comercializado apenas quando satisfaz os critérios de aceitabilidade. A documentação relativa à produção e ao controle de qualidade de cada lote é mantida arquivada
- Para informações de segurança do produto SDS e certificado de análise, visite www.bio-rad.com

Seção 7

Protocolo

Preparação do Meio Desidratado

1. Agite sempre a garrafa antes de usar.
2. Dissolva 43,5 g de pó em 1 L de água destilada e misture até obter uma suspensão homogênea.
3. Aqueça suavemente, agitando com frequência, e depois deixe ferver por menos de 1 minuto.
4. Não prolongue o aquecimento. **Não autoclave.**
5. Resfriar o meio a 50 °C. Dispense em placas de Petri e deixe na bancada para secar.
6. Um frasco de 500 g de pó produz 11,5 L de meio.

Preparação do suplemento seletivo

1. Uma cápsula (3564710) está pronta para uso para 25 g ou 25 ml de amostra diluída em 225 ml de BPW. Tanto a cápsula inteira ou seu conteúdo pode ser adicionado à bolsa de amostras antes da etapa de mistura. A fim de facilitar o manuseio, recomendamos abrir a cápsula e despejar o conteúdo.
2. As cápsulas concentradas (10x) (3564709) devem ser diluídas antes do uso. Coloque o conteúdo de uma cápsula em 10 ml de BPW ou água destilada estéril. Homogenize a fim de obter uma solução concentrada vermelha. Utilize 1 ml de solução concentrada para 25 g ou 25 ml de amostra diluída em 225 ml de BPW. Para volumes grandes de amostra, ajuste o volume do suplemento seletivo para o volume do mix de enriquecimento para obter uma concentração de 4:1000 g/ml. A solução diluída pode ser armazenada por 1 semana à temperatura ambiente ou a 2–8°C.
3. O suplemento em caixa (3564712) é concentrado e deve ser diluído antes do uso. Abra a caixa e encha com 100 ml de BPW ou água destilada estéril. Homogenize agitando vigorosamente a fim de obter uma solução concentrada vermelha. Utilize 1 ml de solução concentrada para 25 g ou 25 ml de amostra diluída em 225 ml de BPW. Para volumes grandes de amostra, ajuste o volume do suplemento seletivo para o volume do mix de enriquecimento para obter uma concentração de 4:1000 g/ml. A solução diluída pode ser armazenada por 1 semana à temperatura ambiente ou a 2–8°C.
4. Abra o frasco do (12013322) e encha com 40 ml de uma solução 1: 1 de etanol / água destilada estéril. Homogeneizar agitando vigorosamente para obter uma solução concentrada azul. Consulte o Guia do usuário do PIF Supplement (documento nº 10000126749) para obter informações sobre o armazenamento da solução diluída.

Enriquecimento da amostra

Escopo (Matrizes)	Enriquecimento	Certificação
Peito de frango cru, ovos, melão e manteiga de amendoim	n g/ml de amostra em 9 x n ml BPW Adicionar RAPID' <i>Salmonella</i> Supplement $41,5 \pm 1^\circ\text{C}$ por 16–22 hr	AOAC
Ampla gama de alimentos, ração animal e amostras ambientais, 25 g	n g/ml de amostra em 9 x n ml BPW Adicionar RAPID' <i>Salmonella</i> Supplement $41,5 \pm 1^\circ\text{C}$ por 16–22 hr	NF VALIDATION
Fórmula infantil e ingredientes lácteos desidratados relacionados, grande protocolo de tamanho de amostra, até 375 g	n g/ml de amostra em 9 x n ml pré-aquecida (37°C) BPW (sem suplemento) $37 \pm 1^\circ\text{C}$ por 20–24 hr	NF VALIDATION
Ampla gama de alimentos, incluindo fórmula infantil com e sem probióticos, ração animal e amostras ambientais	n g/ml de amostra em 9 x n ml BPW Adicionar RAPID' <i>Salmonella</i> Supplement $41,5 \pm 1^\circ\text{C}$ por 16–22 hr	NORDVAL
Produtos lácteos, excluindo leite cru	n g/ml de amostra em 9 x n ml BPW (sem suplemento) $37 \pm 1^\circ\text{C}$ por 16–22 hr Transfira 0,5 ml para 10 ml de RVS pré-aquecido $41,5 \pm 1^\circ\text{C}$ por 6–26 hr	NF VALIDATION
Fórmula infantil e cereais infantis com e sem probióticos e ingredientes, até 375 g	n g/ml de amostra em 3 x n ml pré-aquecida (37°C) BPW Adicionar PIF Supplement $37 \pm 1^\circ\text{C}$ por 18–24 hr	NF VALIDATION
Alimentos para animais de estimação e ração animal, até 375 g	n g/ml de amostra em 5 x n ml pré-aquecida (37°C) BPW Adicionar RAPID' <i>Salmonella</i> Supplement $37 \pm 1^\circ\text{C}$ por 18–24 hr	NF VALIDATION

1. Dilua n g ou n mL de amostra no volume requerido de ATP, seguindo a diluição em razão apropriada. Veja a tabela.
2. Uniformize com um misturador/homogeneizador.
3. Adicione o volume necessário de suplemento seletivo. Volume do suplemento seletivo deve ser adaptado ao volume do mix de enriquecimento (amostra + APT).
4. Incube de acordo com a tabela de Enriquecimento de amostra.
5. Ao seguir o método EN ISO 6579-1, siga o protocolo de enriquecimento de amostras, conforme escrito na norma.
6. Após o passo de enriquecimento, o caldo pode ser armazenado a uma temperatura entre $2\text{--}8^\circ\text{C}$ por 72 h.

Inoculação e Leitura de Meios de Cultura

1. Usando um loop estéril, estriar 10 μl de caldo de enriquecimento no RAPID'*Salmonella* Agar.

2. Incube placas a $37 \pm 1^{\circ}\text{C}$ por 24 ± 2 hr.
3. *Salmonella* spp. formam colônias de cor magenta no RAPID'*Salmonella* Ágar.
4. Após o passo de incubação, as placas podem ser armazenadas a $2-8^{\circ}\text{C}$ por 72 h.

Seção 8

Confirmação de Resultados Positivos

1. No contexto da validação AOAC, verifique as colônias isoladas suspeitas de acordo com o procedimento clássico de teste de confirmação descrito nos métodos de referência padrão.
2. No contexto do NF VALIDATION e NORDVAL, todas as amostras identificadas como positivas devem ser confirmadas de uma das seguintes formas:
 - a. Usando os testes convencionais descritos nos métodos padrão CEN ou ISO (incluindo a etapa de purificação).
 - b. Usar sondas nucleicas, conforme descrito na norma ISO 7218 (por exemplo, kit iQ-Check *Salmonella* spp. PCR em tempo real, nº do catálogo 3578123) usar colônias isoladas (com ou sem etapas de purificação).
 - c. Avaliação da atividade oxidase com um teste de oxidase seguido por um teste Omni-O Omnivalente (A-60) (*Salmonella* Omni-O Antiserum (A-60), nº do catálogo 3560781B) utilizando de uma a três colônias suspeitas isoladas. Se a reação ao teste Omni-O for positiva, proceda com um teste bioquímico ONPG (nº do catálogo 3553822). A *Salmonella* é negativa para o teste de oxidase, positiva para o teste Omni-O (A-60) e negativa para o teste ONPG, com exceção da *Salmonella* positivas para a lactose, que são ONPG positivas.
 - d. Realizar um teste de aglutinação de látex usando o Teste de látex de *Salmonella* (nº do catálogo 3556710) em uma colônia isolada ou usando o teste *Salmonella* Confirm Latex Test (nº do catálogo 3556711) em uma colônia isolada. O *Salmonella* Latex Test de Oxoid também foi validado. A *Salmonella* pertencente aos serogrupos B a E e G são positivos no teste do látex de *Salmonella* (nº do catálogo 3556710).
 - e. Usando o teste MALDI Biotyper (o sistema Bruker Biotyper inclui um espectrômetro de massa MALDI TOF e o software MBT Compass, versão 4) diretamente de uma colônia isolada ou após uma etapa de purificação. As informações de identificação de gênero e/ou espécie fornecidas pela Solução Completa MALDI Biotyper não estão incluídas no escopo da marcação NF VALIDATION.
 - f. Usar qualquer outro método certificado NF VALIDATION baseado em um princípio diferente do da RAPID'*Salmonella* Ágar. O segundo protocolo de método validado deve ser completamente respeitado. Todas as etapas anteriores à etapa de detecção utilizada como ponto de partida para a confirmação devem ser comuns a ambos os métodos.
3. No caso de resultados discordantes (presumivelmente positivos com RAPID'*Salmonella*, negativos com método de confirmação e particularmente com o teste do látex), o laboratório deve seguir as etapas necessárias para garantir a validade dos resultados obtidos.

Seção 9

Confirmação de outros métodos

Confirmação de Resultados Positivos com o kit iQ-Check *Salmonella* II (Protocolo certificado NF VALIDATION)

1. Adicione 500 µl de BPW enriquecido a 10 ml de caldo de RVS pré-aquecido.
2. Incube em $41,5 \pm 1^{\circ}\text{C}$ por 6–26 h.
3. Usando um loop estéril, estriar 10 µl de caldo de enriquecimento em RAPID'*Salmonella* Agar.
4. Incube placas a $37 \pm 1^{\circ}\text{C}$ por 24 ± 2 hr.
5. *Salmonella* spp. formam colônias de cor magenta no RAPID'*Salmonella* Ágar. Realize uma confirmação de uma colônia de magenta usando um teste de aglutinação de látex ou usando os testes convencionais descritos na seção 8.2.

Seção 10

Desempenho e validação do teste

Certificação	Escopo	Protocolo de Validação	Protocolo de Referência	Referência de Certificado
AOAC-RI	Peito de frango cru, ovos, melão e manteiga de amendoim	Métodos de desempenho testados	FDA BAM Cap. 5 USDA MLG 4.10	 Nº da licença 050701
NORDVAL	Ampla gama de alimentos, incluindo fórmula infantil com e sem probióticos, ração animal e amostras ambientais	ISO 16140-2:2016	EN ISO 6579:2002	 NORDVAL # 032
NF VALIDATION	Ampla gama de alimentos, ração animal e amostras ambientais (25 g) Fórmula infantil e ingredientes lácteos desidratados relacionados, grande protocolo de tamanho de amostra (até 375 g), diluição 1:10 Fórmula infantil e cereais infantis com e sem probióticos e ingredientes, grande protocolo de tamanho de amostra (até 375 g), diluição 1:4 Alimentos para animais de estimação e ração animal, grande protocolo de tamanho de amostra (até 375 g), diluição 1:6	EN ISO 16140-2	EN ISO 6579-1	 BRD: 07/11-12/05 MÉTODOS ANALÍTICOS ALTERNATIVOS PARA O AGRONEGÓCIO Certificado pela certificação AFNOR http://nf-validation.afnor.org/en

Seção 11

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United States Department of Agriculture. *Microbiology Laboratory Guidebook*. Chapter 4.10 Isolation and Identification of *Salmonella* from Meat, Poultry Pasteurized Egg, and Siluriformes (Fish) Products and Carcass and Environmental Sponges. <https://www.fsis.usda.gov/wps/wcm/connect/700c05fe-06a2-492a-a6e1-3357f7701f52/mlg-4.pdf?MOD=AJPERES>

United States Food and Drug Administration. Bacteriological Analytical Manual, Chapter 5 *Salmonella*. <http://www.fda.gov/Food/FoodScienceResearch/LaboratoryMethods/ucm070149.htm>

Seção 12

Histórico de Revisão

Data de lançamento	Número do documento	Alteração
Outubro de 2020	10000126748 Ver A	- Alteração importante - Novo design de documento - Alteração do número d documento – versão anterior RAPID_Salmonella_V11_25 de junho de 2018
Junho de 2021	10000126748 Ver B	Suplemento em novo formato
Janeiro de 2023	10000126748 Ver C	- Retirada do formato de suplemento em comprimido - Atualização do preparo do suplemento seletivo - Extensão da validação NF para alimentos de animais de estimação e ração animal até 375 g

Visite www.bio-rad.com/rapidmedia para obter mais informações sobre a nossa completa linha de meios cromogênicos RAPID.

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10000126748 Ver C US/EG

RAPID'*Salmonella* Agar

Manual del usuario

Medio cromogénico para la detección de *Salmonella* spp. en productos alimentarios destinados al consumo humano, consumo animal y en muestras ambientales

Referencia #3563961, placas preparadas, placas de 90 mm x 20
Referencia #3563963, placas preparadas, placas de 90 mm x 120
Referencia #3564705, deshidratado, 500 g



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Tabla de Contenidos

Apartado 1	Introducción	1
Apartado 2	Principio de RAPID' <i>Salmonella</i>	1
Apartado 3	Fórmula teórica	1
Apartado 4	Vida útil y conservación.....	1
Apartado 5	Materiales necesarios, pero no suministrados.....	2
	Equipamiento	2
	Fungibles	2
Apartado 6	Precauciones, limitaciones de uso y control de calidad.....	2
Apartado 7	Protocolo	3
	Preparación del medio deshidratado	3
	Preparación del suplemento selectivo	4
	Enriquecimiento de la muestra.....	4
	Siembra y lectura de la placa	5
Apartado 8	Confirmación de los resultados positivos.....	5
Apartado 9	Confirmación de otros métodos	6
Apartado 10	Aplicaciones del ensayo y validaciones	7
Apartado 11	Referencias	7
Apartado 12	Historial de revisiones	8

Apartado 1 Introducción

La *Salmonella* es una bacteria Gram-negativa, que no forma esporas, con forma de bacilo y que es predominantemente móvil, con sólo unas pocas excepciones. La *Salmonella* ha sido una de las dos causas más frecuentes de enfermedades bacterianas transmitidas por los alimentos. Los síntomas comunes de la salmonelosis incluyen diarrea, fiebre y calambres abdominales que se presentan de 12 a 72 hr después de la infección. En estudios se ha demostrado que por cada caso de *Salmonella* confirmado en laboratorio, se originan 38 casos en la comunidad que no se notifican. Los análisis para detectar la presencia de *Salmonella* en los alimentos y en el entorno de procesamiento representan una práctica importante. El uso de sustratos cromogénicos en los medios de cultivo facilita el uso de los mismos con resultados basados en reacciones enzimáticas de cambio de color.

Apartado 2 Principio de RAPID'*Salmonella*

RAPID'*Salmonella* Agar permite la identificación presuntiva de *Salmonella* spp. móvil y no móvil mediante la detección de la actividad C8-esterasa. El cribado simultáneo de la actividad β -glucosidasa permite diferenciar las colonias de *Salmonella* de las de otras *Enterobacteriaceae*. Después de la incubación, *Salmonella* aparece como colonias fácilmente identificables, típicamente de color magenta, mientras que las bacterias que no son *Salmonella* crecen como colonias azules o incoloras. RAPID'*Salmonella* Agar permite la detección de *Salmonella* móvil y no móvil, así como de *Salmonella* Typhi y *Salmonella* Paratyphi lactosa-positivas.

Apartado 3 Fórmula teórica

Mezcla nutritiva	14,5 g
Agentes selectivos	14 g
Mezcla cromogénica	2,3 g
Agar	12,7 g
Agua destilada	c.s.p. 1.000 ml

pH final a 25° C = 7,2 $\pm$ 0,2

Apartado 4 Vida útil y conservación

- Agar deshidratado: a 2–8° C en un envase cuidadosamente sellado, en un lugar seco y oscuro
- Agar preparado: 2–8° C en un lugar oscuro
- Placa preparada desde agar deshidratado: 15 días a 2–8° C en un envase cuidadosamente sellado, en un lugar seco y oscuro

Apartado 5

Materiales necesarios, pero no suministrados

Equipamiento

- Todo el instrumental habitual en laboratorio
- Placa calefactada
- Balanza, sensibilidad de 0,1 g
- Agitador/homogeneizador
- Incubador o sala de incubación controlada termostáticamente, con una precisión de $\pm 1^\circ \text{C}$
- Baño termostático

Fungibles

- Caldo de enriquecimiento: Agua de peptona tamponada (APT) (referencia #3554179, 225 ml x 6 frascos; 3564684, deshidratado 500 g; 3555790, 5 L x 2 bolsas; 3555795, 3 L x 4 bolsas)
- Asas de siembra
- PIF Supplement (referencia #12013322, 2 g)
- RAPID'*Salmonella* suplemento selectivo para protocolo corto (referencia #3564709, 10x concentrado x 100 cápsulas; 3564710, 1x concentrado x 100 cápsulas; 3564712, 1 caja x 100 pruebas)
- Rappaport Vassiliadis Soja (RVS) para el protocolo de doble enriquecimiento (referencia #3555773, 10 ml x 25 tubos; 3564324, deshidratado 500 g)
- Placas de Petri estériles
- Pipetas estériles
- Bolsas estériles para pesada

Apartado 6

Precauciones, limitaciones de uso y control de calidad

Precauciones

- Deben respetarse las buenas prácticas de laboratorio (EN ISO 7218). Se debe usar una protección adecuada, como guantes y batas de laboratorio, cuando se trabaja con bacterias vivas potencialmente infecciosas como *Salmonella*.
- Los medios que han estado en contacto con muestras de alimentos deben considerarse potencialmente contaminados y deben eliminarse de conformidad con las normas y reglamentos locales.

- Si se añade la cápsula de suplemento selectivo al agua de peptona tamponada (APT), hay que utilizar unas pinzas estériles para añadirla a la bolsa. (Recomendamos que compruebe la total apertura de la cápsula durante la etapa de homogeneización en estomacher.) Si se manipula el contenido de la cápsula con los dedos, no se debe añadir al caldo de enriquecimiento debido al potencial riesgo de contaminación.
- Las cápsulas RAPID'*Salmonella* y las tabletas RAPID'*Salmonella* contienen agentes selectivos y un excipiente. Los agentes selectivos se disuelven muy bien. Sin embargo, el excipiente permanece en suspensión y puede formar un depósito cuando el contenido de la cápsula se diluye en una pequeña cantidad de agua de peptona tamponada (APT) o agua destilada estéril. Por lo tanto, siempre se debe agitar bien antes de usar la solución concentrada.

Limitaciones de uso

- La prueba ONPG (β -galactosidasa) excluye la confirmación de *Salmonella* lactosa-positiva.
- Aunque las cepas de *Salmonella* con mayor prevalencia pueden ser detectadas por *Salmonella* Confirm Latex Kit, el kit no detectó 41 de las 150 cepas analizadas durante la extensión de la Validación NF de RAPID'*Salmonella*.
- Ciertas cepas de *Salmonella* (algunas forman parte de la *Salmonella* serovar Dublín y de las especies de *S. bongori*) pueden manifestar un color magenta débil debido a una baja actividad de la esterasa.
- En el contexto de la validación de NORDVAL, el protocolo de doble enriquecimiento fue validado con un tiempo de enriquecimiento en RVS de sólo 24 ± 2 hr.
- Tras la etapa de incubación, la placa puede almacenarse a $2-8^{\circ}\text{C}$ durante 72 hr. Algunas cepas que no son de *Salmonella* (por ejemplo, *E. coli*) pueden desarrollar colonias de color rosado después de este almacenamiento. Estas colonias dudosas muestran una prueba de confirmación negativa.

Control de calidad

- Todos los productos fabricados y comercializados por Bio-Rad están sujetos a un protocolo de garantía de calidad en todas las etapas, desde la recepción de las materias primas hasta la comercialización de los productos acabados. Cada lote de producto acabado se somete a un control de calidad según la norma EN ISO 11133 y su comercialización está condicionada a que cumpla los criterios de aceptabilidad. La documentación relativa a la producción y el control de calidad de cada lote se mantiene archivada.
- Para información relativa a la seguridad sobre el producto (SDS) y del certificado de análisis, visite www.bio-rad.com

Apartado 7 Protocolo

Preparación del medio deshidratado

1. Agitar siempre el frasco antes de usar.
2. Disolver 43,5 g de polvo en 1 L de agua destilada y mezclar hasta obtener una suspensión homogénea.

3. Calentar suavemente, agitando frecuentemente, y luego llevar a ebullición sin exceder 1 minuto.
4. No prolongar el calentamiento. **No utilizar el autoclave.**
5. Enfriar el medio a 50° C. Dispensar en placas de Petri y dejar en una superficie nivelada para que se sequen.
6. Un frasco de 500 g de medio deshidratado permite obtener 11,5 L de medio.

Preparación del suplemento selectivo

1. Una cápsula (3564710) está lista para ser usada para 25 g o 25 ml de muestra diluida en 225 ml de APT. Tanto la cápsula entera como su contenido pueden añadirse a la bolsa de muestra antes del paso de homogeneización. Para facilitar la manipulación, recomendamos abrir la cápsula y verter el contenido.
2. Las cápsulas concentradas (10x) (3564709) deben ser diluidas antes de su uso. Vierta el contenido de una cápsula en 10 ml de APT o agua destilada estéril. Homogeneice para obtener una solución concentrada de color rojo. Use 1 ml de solución concentrada para 25 g o 25 ml de muestra diluida en 225 ml de APT. Para muestras de tamaño elevado, ajuste el volumen de suplemento al volumen de la mezcla de enriquecimiento, para obtener una concentración de 4:1000 g/ml. La solución diluida puede conservarse durante 1 semana a temperatura ambiente o a 2-8° C.
3. El suplemento en frasco (3564712) es concentrado y debe ser diluido antes de su uso. Abra el frasco y adicione 100 ml de APT o agua destilada estéril. Homogeneice agitando enérgicamente para obtener una solución concentrada roja. Use 1 ml de solución concentrada para 25 g o 25 ml de muestra diluida en 225 ml de APT. Para muestras de tamaño elevado, ajuste el volumen de suplemento al volumen de la mezcla de enriquecimiento, para obtener una concentración de 4:1000 g/ml. La solución diluida puede conservarse durante 1 semana a temperatura ambiente o a 2-8° C.
4. Abra la botella de PIF Supplement (12013322) y agregue 40 ml de solución 1:1 etanol/agua destilada. Homogenice agitando vigorosamente hasta obtener una solución azul concentrada. Consulte la guía de usuario PIF Supplement (documento #10000126749) para información sobre condiciones de almacenamiento y dilución de la solución.

Enriquecimiento de la muestra

Alcance (matrices)	Enriquecimiento	Certificación
Pechuga de pollo cruda, huevos, melón y mantequilla de cacahuete.	<i>n</i> g/ml de muestra en 9 x <i>n</i> ml de APT Agregue RAPID'Salmonella Supplement 41,5 ± 1° C durante 16–22 hr	AOAC
Amplia gama de alimentos de consumo humano, alimentos de consumo animal y muestras ambientales, 25 g	<i>n</i> g/ml de muestra en 9 x <i>n</i> ml de APT Agregue RAPID'Salmonella Supplement 41,5 ± 1° C durante 16–22 hr	NF VALIDATION
Fórmula infantil e ingredientes lácteos deshidratados relacionados, protocolo para tamaño de muestra grande, hasta 375 g	<i>n</i> g/ml de muestra en 9 x <i>n</i> ml de APT precalentada (37° C) (sin suplemento) 37 ± 1° C durante 20–24 hr	NF VALIDATION
Amplia gama de alimentos, incluyendo fórmulas infantiles con y sin probióticos, alimentos para animales y muestras ambientales.	<i>n</i> g/ml de muestra en 9 x <i>n</i> ml de APT Agregue RAPID'Salmonella Supplement 41,5 ± 1° C durante 16–22 hr	NORDVAL
Productos lácteos, excluyendo la leche	<i>n</i> g/ml de muestra en 9 x <i>n</i> ml de APT	NF VALIDATION

Apartado 8 Confirmación de los resultados positivos

cruda	(sin suplemento) 37 ± 1° C durante 16–22 hr Transferir 0,5 ml a 10 ml de RVS precalentado 41,5 ± 1° C durante 6–26 hr	
Fórmula infantil y cereales para infantes con y sin probióticos e ingredientes, hasta 375 g	<i>n</i> g/ml de muestra en 3 x <i>n</i> ml de APT precalentada (37° C) Agregue PIF Supplement 37 ± 1° C durante 18–24 hr	NF VALIDATION
Comida para mascotas y alimentación animal, hasta 375 g	<i>n</i> g/ml de muestra en 5 x <i>n</i> ml de APT precalentada (37° C) Agregue RAPID'Salmonella Supplement 37 ± 1° C durante 18–24 hr	NF VALIDATION

1. Diluir *n* g o *n* ml de muestra en el volumen requerido de APT siguiendo la proporción de dilución adecuada. Véase la tabla.
2. Homogeneizar con agitador/homogeneizador.
3. Añadir el volumen requerido de suplemento selectivo. El volumen de suplemento selectivo debe adaptarse al volumen de la mezcla de enriquecimiento (muestra + BPW).
4. Incubar conforme a la tabla de enriquecimiento según el tipo de muestra.
5. Cuando se aplica el método EN ISO 6579-1, debe seguirse el protocolo de enriquecimiento de la muestra tal como se establece en la norma.
6. Después de la etapa de enriquecimiento, el caldo se puede conservar a 2–8 °C durante 72 hr.

Siembra y lectura de la placa

1. Usando un asa estéril, siembre 10 µl de caldo de enriquecimiento en RAPID'Salmonella Agar.
2. Incube las placas a 37 ± 1° C durante 24 ± 2 hr.
3. *Salmonella* spp. forma colonias de color magenta en RAPID'Salmonella Agar.
4. Tras la etapa de incubación, las placas pueden almacenarse a 2-8° C durante 72 hr.

Apartado 8 Confirmación de los resultados positivos

1. En el contexto de la validación AOAC, confirme las colonias sospechosas aisladas siguiendo el procedimiento clásico de confirmación indicado en los métodos de referencia normalizados.
2. Con respecto a la validación NF VALIDATION y NORDVAL, todas las muestras identificadas como positivas deben confirmarse de una de las siguientes formas:
 - a. Empleando las pruebas convencionales descritas en los métodos normalizados CEN o ISO (incluida la fase de purificación).
 - b. Utilizando sondas nucleicas como se describe en la norma ISO 7218 (por ejemplo, iQ-Check *Salmonella* II Real-Time PCR Kit, referencia #3578123) sobre colonias aisladas (con o sin fase de purificación).

- c. Evaluando la actividad de la oxidasa con una prueba de oxidasa seguida de un test omnivalente Omni-O (A-60) (Antisuero *Salmonella* Omni-O (A-60) referencia #3560781B) utilizando de una a tres colonias sospechosas aisladas. Si la reacción a la prueba de aglutinación de Omni-O resulta positiva, proceda con una prueba bioquímica de ONPG (referencia #3553822). La *Salmonella* resulta negativa en la prueba de la oxidasa, positiva en la prueba de Omni-O (A-60), y negativa en la prueba de ONPG, con excepción de *Salmonella* lactosa-positiva, que es positiva para ONPG.
 - d. Realizando una prueba de aglutinación de látex como *Salmonella* Latex Test (referencia #3556710) o *Salmonella* Confirm Latex Test (referencia #3556711) sobre una colonia aislada. También se ha validado el látex de *Salmonella* de Oxoid. Las *Salmonellas* pertenecientes a los serogrupos B al E y G son positivas en la prueba de aglutinación de látex como *Salmonella* Latex Test (referencia #3556710).
 - e. Usando MALDI Biotyper test (sistema de biotipado de Bruker que incluye un espectrofotómetro de masas MALDI time-of-flight y MBT Compass Software, version 4) directamente sobre una colonia aislada o tras una fase de purificación. La información sobre identificación de género y/o especie proporcionada por la solución completa MALDI Biotyper no está incluida en el alcance de NF VALIDATION.
 - f. Usando cualquier otro método certificado NF VALIDATION basado en un principio diferente al de RAPID'*Salmonella* Agar. El protocolo validado de este segundo método debe ser seguido en su totalidad. Todos los pasos que preceden al paso de detección utilizado como punto de partida para la confirmación deben ser comunes para ambos métodos.
3. En caso de producirse resultados discordantes (supuestos positivos con RAPID'*Salmonella*, negativos con el método de confirmación, y en particular con la prueba de látex), el laboratorio debe seguir los pasos necesarios para garantizar la validez de los resultados obtenidos.

Apartado 9

Confirmación de otros métodos

Confirmación de los resultados positivos con iQ-Check *Salmonella* II Kit (protocolo certificado NF VALIDATION)

1. Añadir 500 µl de agua de peptona tamponada (APT) en 10 ml de caldo RVS precalentado.
2. Incubar a $41,5 \pm 1^\circ \text{C}$ durante 6–26 hr.
3. Usando un asa estéril, aislar 10 µl de caldo de enriquecimiento en RAPID'*Salmonella* Agar.
4. Incubar las placas a $37 \pm 1^\circ \text{C}$ durante 24 ± 2 hr.
5. *Salmonella* spp. forma colonias de color magenta en RAPID'*Salmonella* Agar. Realice la confirmación de una colonia característica mediante una prueba de aglutinación de látex o mediante las pruebas convencionales descritas en el apartado 8.2.

Apartado 10

Aplicaciones del ensayo y validaciones

Certificación	Alcance	Protocolo de validación	Protocolo de referencia	Referencia de certificado
AOAC-RI	Pechuga de pollo cruda, huevos, melón y mantequilla de cacahuete.	Performance Tested Methods	FDA BAM Chap. 5 USDA MLG 4.10	 Licencia# 050701
NORDVAL	Amplia gama de alimentos, incluyendo fórmulas infantiles con y sin probióticos, alimentos para animales y muestras ambientales.	ISO 16140-2:2016	EN ISO 6579:2002	 NORDVAL # 032
NF VALIDATION	Amplia gama de alimentos, piensos y muestras ambientales (25 g) Fórmulas infantiles e ingredientes lácteos deshidratados relacionados, protocolo para tamaño de muestra grande, hasta 375 g, dilución 1:10 Fórmula infantil y cereales para infantes con y sin probióticos e ingredientes, protocolo para tamaño de muestra grande, hasta 375 g, dilución 1:4 Comida para mascotas y alimentación animal, protocolo para tamaño de muestra grande, hasta 375 g, dilución 1:6	EN ISO 16140-2	EN ISO 6579-1	 BRD: 07/11–12/05 MÉTODOS ANALÍTICOS ALTERNATIVOS PARA LA AGROINDUSTRIA Certificado mediante certificación AFNOR http://nf-validation.afnor.org/en

Apartado 11

Referencias

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ISO 6579-1: 2017. Microbiología de la cadena alimentaria — Método horizontal para la detección, recuento y serotipado de *Salmonella* — Parte 1: Detección de *Salmonella* spp.

Manafi M et al. (1991). Fluorogenic and chromogenic substrates used in bacterial diagnostics. Microbiol Rev, 55, 335–348.

Perry JD et al. (1999). ABC medium, a new chromogenic agar for selective isolation of *Salmonella* spp. J Clin Microbiol, 37, 766–768.

United States Department of Agriculture. *Microbiology Laboratory Guidebook*. Chapter 4.10 Isolation and Identification of *Salmonella* from Meat, Poultry Pasteurized Egg, and Siluriformes (Fish) Products and Carcass and Environmental Sponges. <https://www.fsis.usda.gov/wps/wcm/connect/700c05fe-06a2-492a-a6e1-3357f7701f52/mlg-4.pdf?MOD=AJPERES>

United States Food and Drug Administration. Bacteriological Analytical Manual, Chapter 5 *Salmonella*. <http://www.fda.gov/Food/FoodScienceResearch/LaboratoryMethods/ucm070149.htm>

Apartado 12

Historial de revisiones

Fecha de publicación	N.º de documento	Cambio
Octubre de 2020	10000126748 Ver A	- Cambio significativo - Nuevo diseño del documento - Cambio en el número de documento – versión anterior RAPID_Salmonella_V11_25 junio 2018
Junio de 2021	10000126748 Ver B	Suplemento en nuevo formato
Enero de 2023	10000126748 Ver C	- Retirada del suplemento en formato de pastilla - Actualización de la preparación del suplemento - Extensión de la Validación NF en comida para mascotas y alimentación animal hasta 375 g

Visite www.bio-rad.com/rapidmedia para más información sobre nuestra gama completa de medios cromogénicos

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10000126748 Ver C US/EG

RAPID'*Salmonella* Agar

用户指南

用于检测供人类食用的食品、动物饲料和环境样品中的沙门氏菌属的显色培养基

目录 #3563961, 即用型平板, 90 mm x 20 个培养皿

目录 #3563963, 即用型平板, 90 mm x 120 个培养皿

目录 #3564705, 干粉, 500 g



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目录

第 1 部分 简介.....	1
第 2 部分 RAPID' <i>Salmonella</i> 原理.....	1
第 3 部分 理论配方	1
第 4 部分 保质期及储存条件	1
第 5 部分 其他仪器、试剂与耗材	2
仪器	2
试剂和耗材.....	2
第 6 部分 预防措施、使用限制和质量控制	2
第 7 部分 操作流程.....	3
干粉培养基的制备.....	3
选择性添加剂的制备.....	3
样品增菌	4
接种和平板读数.....	5
第 8 部分 阳性结果的确认	5
第 9 部分 其他方法的确认	6
第 10 部分 测试性能和验证.....	6
第 11 部分 参考资料.....	7
第 12 部分 修订记录.....	8

第 1 部分 简介

第 1 部分

简介

*沙门氏菌*是一种革兰氏阴性的无芽孢杆菌，主要为运动型细菌，只有少数例外。*沙门氏菌*是导致细菌性食源性疾病的两种最常见原因之一。*沙门氏菌*病的常见症状包括感染后 12-72 小时出现的腹泻、发烧和腹部绞痛。研究表明，每一个实验室确诊的*沙门氏菌*病例，在社区就有 38 例未上报。检测食品和加工环境中是否存在*沙门氏菌*是一项重要的规定。在培养基中使用显色底物提高了培养基的易用性，根据酶促反应引起的显著颜色变化从而得出最终结果。

第 2 部分

RAPID' *Salmonella* 原理

RAPID' *Salmonella* 平板可以通过检测 C8-酯酶活性来推定鉴定运动型和非运动型*沙门氏菌*。同时筛选 β -葡萄糖苷酶的活性可以将*沙门氏菌*菌落与其他*肠杆菌科*的菌落区分开来。经培养后，*沙门氏菌*显示为容易识别、典型的紫色菌落，而非*沙门氏菌*则显示蓝色或无色的菌落。RAPID' *Salmonella* 平板可以检测运动型和非运动型的*沙门氏菌*，以及乳糖阳性的伤寒*沙门氏菌*和副伤寒*沙门氏菌*。

第 3 部分

理论配方

营养混合物	14.5 g
选择性制剂	14 g
显色混合物	2.3 g
琼脂	12.7 g
蒸馏水	qsp 1,000 ml

25°C 时的最终 pH 值 = 7.2 $\pm$ 0.2

第 4 部分

保质期及储存条件

- 干粉培养基：在 2–8° C 下妥善密封包装，储存在干燥避光处
- 预备培养基：2–8° C 避光处
- 干粉平板的制备：在 2–8° C 下妥善密封包装，置于干燥避光处储存 15 天

第 5 部分

其他仪器、试剂与耗材

仪器

- 所有常用的实验室仪器
- 热平板
- 天平，灵敏度为 0.1 g
- 搅拌器/均质器
- 恒温控制的孵化器或孵化室，精确到 $\pm 1^{\circ}\text{C}$
- 水浴

试剂和耗材

- 培养基：缓冲蛋白胨水 (BPW) (目录 #3554179, 225 ml x 6 瓶；3564684, 干粉 500 g；3555790, 5 L x 2 袋；3555795, 3 L x 4 袋)
- 接种环
- PIF Supplement (目录 #12013322, 2 g 瓶)
- RAPID'*Salmonella* selective supplement for short protocol
(用于简要方案的 RAPID'*Salmonella* 选择性添加剂, 目录 #3564709, 10x 浓缩 x 100 粒；#3564710, 1x 浓缩 x 100 粒；#3564712, 1 盒 x 100 次试验)
- 用于两步增菌方案的 Rappaport Vassiliadis Soy (RVS) (目录 #3555773, 10 ml x 25 管 x 25；3564324, 干粉 500 g)
- 无菌培养皿
- 无菌移液管
- 无菌称重袋

第 6 部分

预防措施、使用限制和质量控制

预防措施

- 遵守良好实验室规范 (EN ISO 7218)。在处理沙门氏菌等具有潜在传染性的活细菌时，应穿戴适当的防护装置，如手套和实验室外套
- 与食品样品接触过的培养基应被视为潜在传染性材料处理，并根据当地法规 and 规定进行废弃物处理

第 7 部分

操作流程

- 如果在 BPW 中加入整粒选择性添加剂胶囊，必须使用无菌镊子将胶囊加入袋中。（我们建议您检查胶囊是否实际在胃化阶段打开。）如果手指不小心碰到胶囊内物质，由于有污染的风险，不能将其加入增菌液里
- RAPID'*Salmonella* 胶囊和 RAPID'*Salmonella* 添加剂含有选择性试剂和赋形剂。选择性试剂溶解得很好。然而，当胶囊内物质在少量 BPW 或无菌蒸馏水中稀释时，赋形剂保持悬浮状态并可能产生沉淀。因此，在使用浓缩溶液之前，一定要充分摇晃。

使用限制

- ONPG 确认试验排除了乳糖阳性沙门氏菌的确认
- 尽管 *Salmonella* Confirm Latex Kit 可以检测到最普遍的沙门氏菌菌株，但在 RAPID'*Salmonella* 的 NF 验证扩展试验中，该试剂盒未检测到 150 种菌株中的 41 种菌株
- 一些沙门氏菌菌株（少数属于都柏林沙门氏菌血清型和邦戈里沙门氏菌）由于酯酶活性低，可以显示出微弱的紫色
- 在 NORDVAL 验证的背景下，两步法增菌方案进行了验证，RVS 增菌时间仅需 24 ± 2 小时
- 培养结束后，平板可以在 $2-8^{\circ}\text{C}$ 下储存 72 小时。一些非沙门氏菌菌株（例如，大肠杆菌）在这种储存方式后可形成粉红色的菌落。这些可疑菌落的确认测试为阴性

质量控制

- Bio-Rad 公司生产和销售的每一种产品，从接收原材料到销售成品的各个阶段都要受到质量保证程序的约束。每批成品都根据 EN ISO 11133 进行质量控制，只有满足验收标准才能上市。与每批次的质量和um控制有关的文件均进行存档
- 有关 SDS 产品安全信息和分析证书，请访问 www.bio-rad.com。

第 7 部分

操作流程

干粉培养基的制备

1. 使用前请摇晃瓶子。
2. 将 43.5 g 粉末溶解在 1 L 蒸馏水中，并进行搅拌，直到得到均匀的悬浮液。
3. 缓慢加热，不断搅拌，然后煮沸少于 1 分钟。
4. 请勿长时间加热。**请勿进行高压灭菌。**
5. 将培养基冷却到 50°C 。分装在培养皿中，放在台面上晾干。
6. 一瓶 500 g 粉末可制成 11.5 L 培养基。

选择性添加剂的制备

1. 一粒胶囊 (3564710) 可用于 25 g 或 25 ml 的样品稀释在 225 ml 的 BPW 中，即可使用。在混合步骤之前，可以将整个胶囊或其内容物加入样品袋中。为了便于操作，我们建议打开胶囊并倒出所有物质。
2. 浓缩 (10x) 胶囊 (3564709) 在使用前必须稀释。将一粒胶囊的内容物倒入 10 ml BPW 或无菌蒸馏水中。均质化以获得红色浓缩溶液。使用 1 ml 浓缩溶液处理 25 g 或 25 ml 样品稀释在 225 ml BPW 中。对于大样本量，调整选择性添加剂和增菌混合液的体积以获得 4:1000 g/ml 的浓度。稀释后的溶液可在环境温度下或 2-8°C 下储存 1 周。
3. 盒装添加剂 (3564712) 是浓缩的，使用前必须稀释。打开盒子，装入 100 ml BPW 或无菌蒸馏水。通过剧烈搅拌使其均匀，以获得红色的浓缩溶液。使用 1 ml 浓缩溶液处理 25 g 或 25 ml 样品稀释在 225 ml BPW 中。对于大样本量，调整选择性添加剂和增菌混合液的体积以获得 4:1000 g/ml 的浓度。稀释后的溶液可在环境温度下或 2-8°C 下储存 1 周。
4. 打开 PIF 添加剂 (12013322) 的瓶子，加入 40 ml 1:1 乙醇/无菌蒸馏水溶液。通过剧烈搅拌使其均匀，以获得蓝色的浓缩溶液。有关稀释液储存的信息，请参阅 PIF 添加剂用户指南 (文档 #10000126749)。

样品增菌

范围（食品基质）	增菌条件	证书
生鸡胸肉、鸡蛋、哈密瓜和花生酱	n g/ml 样品在 $9 \times n$ ml BPW 中 加入 RAPID' <i>Salmonella</i> 添加剂 $41.5 \pm 1^\circ \text{C}$ 下培养 16–22 小时	AOAC
广泛的食物、动物饲料和环境样品，25 g	n g/ml 样品在 $9 \times n$ ml BPW 中 加入 RAPID' <i>Salmonella</i> 添加剂 $41.5 \pm 1^\circ \text{C}$ 下培养 16–22 小时	NF 验证
婴儿配方奶粉和相关干粉乳制品成分， 大样本方案，料最多 375 g	n g/ml 样品在 $9 \times n$ ml 预热 (37°C) BPW 中（无添加剂） $37 \pm 1^\circ \text{C}$ 下培养 20–24 小时	NF 验证
广泛的食物，包括有或无益生菌的婴儿配 方奶粉、动物饲料和环境样品	n g/ml 样品在 $9 \times n$ ml BPW 中 加入 RAPID' <i>Salmonella</i> 添加剂 $41.5 \pm 1^\circ \text{C}$ 下培养 16–22 小时	NORDVAL
乳制品，不包括生牛奶	n g/ml 样品在 $9 \times n$ ml BPW 中（无添加剂） $37 \pm 1^\circ \text{C}$ 下培养 16–22 小时 将 0.5 ml 转移到 10 ml 预热的 RVS $41.5 \pm 1^\circ \text{C}$ 下培养 6–26 小时	NF 验证
有或无益生菌和成分的婴儿配方奶粉和婴 儿谷物食品，料最多 375 g	n g 样品在 $3 \times n$ ml 预热 (37°C) BPW 中 加入 PIF 添加剂 $37 \pm 1^\circ \text{C}$ 下培养 18–24 小时	NF 验证
宠物食品及动物饲料最多 375 g	n g 样品在 $5 \times n$ ml 预热 (37°C) BPW 中	NF 验证

第 8 部分

阳性结果的确认

	加入 RAPID' <i>Salmonella</i> 添加剂 37 ± 1° C 下培养 18-24 小时	
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1. 按照适当的稀释比例，在所需体积的 BPW 中稀释 ng 或 nml 样品。见表。
2. 用搅拌器/均质器进行均质处理。
3. 加入所需量的选择性添加剂。调整选择性添加剂与增菌混合液的体积（样品 + BPW）。
4. 根据样品增菌表进行培养。
5. 当遵循 EN ISO 6579-1 方法时，请遵循标准中写明的样品增菌方案。
6. 增菌步骤后，增菌液可在 2-8°C 下储存 72 小时。

接种和平板读数

1. 使用无菌环，在 RAPID' *Salmonella* 平板上滴加 10 μl 增菌液。
2. 在 37 ± 1°C 下培养平板 24 ± 2 小时。
3. 沙门氏菌属在 RAPID' *Salmonella* 平板上形成紫色菌落。
4. 培养结束后，平板可以在 2-8° C 下储存 72 小时。

第 8 部分

阳性结果的确认

1. 在 AOAC 验证的背景下，根据标准参考方法中描述的经典确认测试程序确认可疑的分离菌落。
2. 在 NF 验证和 NORDVAL 的背景下，所有被确定为阳性的样品必须以下列方式之一加以确认：
 - a. 使用 CEN 或 ISO 标准方法中描述的常规测试（包括纯化步骤）。
 - b. 使用 ISO 7218 标准中描述的核探针（例如 iQ-Check *Salmonella* II Real-Time PCR Kit，目录 #3578123）、使用分离的菌落（有或无纯化步骤）。
 - c. 使用一到三个分离的可疑菌落，通过氧化酶试验和 Omnivalent Omni-O 试验 (A-60)（*Salmonella* Omni-O Antiserum (A-60) 目录 #3560781）评估氧化酶活性。如果对 Omni-O 试验的反应呈阳性，则继续进行 ONPG 生化试验（目录 #3553822）。沙门氏菌对氧化酶试验呈阴性，对 Omni-O 试验 (A-60) 呈阳性，对 ONPG 试验呈阴性，但乳糖阳性的沙门氏菌除外，它们是 ONPG 阳性。
 - d. 使用 *Salmonella* Latex Test（目录 #3556710）对分离的菌落进行乳胶凝集试验，或使用 *Salmonella* Confirm Latex Test（目录 #3556711）对分离的菌落进行试验。Oxoid 的 *Salmonella* Latex Test 也已得到验证。属于 B 至 E 和 G 血清群的沙门氏菌在乳胶试验中呈阳性 *Salmonella* Latex Test（目录 #3556710）。

- e. 直接从分离的菌落或在纯化步骤之后采用 MALDI Biotyper 试验（Bruker Biotyper 系统包括 MALDI 飞行时间质谱仪和 MBT Compass 软件，第 4 版）。由 MALDI Biotyper Complete Solution 提供的菌属和/或菌种的鉴定信息不包括在 NF 验证标志的范围内。
 - f. 使用任何其他与 RAPID' *Salmonella* 平板不同原理的 NF 验证认证的方法。必须完全遵循第二种方法的验证方案。用作确认起点的检测步骤之前的所有步骤必须对两种方法通用。
3. 如果出现不一致的结果（RAPID' *Salmonella* 推定为阳性，而确认法特别是乳胶试验则为阴性），实验室必须遵循必要的步骤，以确保所获结果的有效性。

第 9 部分

其他方法的确认

用 iQ-Check *Salmonella* II Kit 确认阳性结果（NF 验证认证方案）

1. 将 500 µl 增菌 BPW 加入到 10 ml 预热的 RVS 肉汤中。
2. 在 $41.5 \pm 1^\circ \text{C}$ 下培养 6–26 小时。
3. 使用无菌环，在 RAPID' *Salmonella* 平板上滴加 10 µl 增菌液。
4. 在 $37 \pm 1^\circ \text{C}$ 下培养平板 24 ± 2 小时。
5. 沙门氏菌属在 RAPID' *Salmonella* 平板上形成紫色菌落。使用乳胶凝集试验或使用第 8.2 部分所述的常规试验对一个品红色菌落进行确认。

第 10 部分

测试性能和验证

证书	范围	验证方案	参考方案	证书参考
AOAC-RI	生鸡胸肉、鸡蛋、 哈密瓜和花生酱	性能测试方法	FDA BAM 第 5 章 USDA MLG 4.10	 许可 # 050701
NORDVAL	广泛的食物，包括有或无益生 菌的婴儿配方奶粉、 动物饲料和环境样品	ISO 16140- 2:2016	EN ISO 6579:2002	

				NORDVAL # 032
NF 验证	广泛的食物、动物饲料和环境样品 (25 g) 婴儿配方奶粉和相关干粉乳制品成分, 大样本方案, (料最多375 g), 1:10 稀释比 有或无益生菌和成分的婴儿配方奶粉和婴儿谷物食品, 大样本方案 (料最多 375 g), 1:4 稀释比 宠物食品及动物饲大样本方案 (料最多 375 g), 1:6 稀释比	EN ISO 16140-2	EN ISO 6579-1	 BRD : 07/11-12/05 农业企业的替代分析方法 通过 AFNOR 认证 http://nf-validation.afnor.org/en

第 11 部分

参考资料

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第 12 部分

修订记录

发布日期	文件编号	变更
2020 年 10 月	10000126748 Ver A	- 扩展了 1:4 稀释的 PIF 添加剂的大样本量的验证 - 新的文档设计 - 文件编号变更 – 先前版本 RAPID_Salmonella_V11_25 2018 年 6 月
2021 年 6 月	10000126748 Ver B	新的补充格式
2023 年 1 月	10000126748 Ver C	- 撤回添加剂片剂形式 - 选择性添加剂制备更新 - 宠物食品及动物饲料的 NF 验证扩展至 375 克

请访问 www.bio-rad.com/rapidmedia, 了解有关我们全系列 RAPID 显色培养基的更多信息。

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